The two forms of the self-interaction on the 4+4 spacetime: definitions, consequences, discussion

PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham_definition_C — provenance P3: the covariant form (the definition in this folder) and the quantized-sector form (the alternative), recorded completely

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0. What this document is

The 16-component Dirac field of PROJECT spacetimes has a mass term and a quartic self-interaction with nine couplings $g_0, \dots, g_8$. On the spacetime with four space-like and four time-like coordinates (the **4+4 spacetime**) the quartic term can be written in two ways that are both natural and that are **not equivalent**. They are called here

- **form C**, the *covariant form*: the nine invariants are built with the adjoint $\tilde{\psi} = \psi^\dagger C^-$ that makes them invariant under all rotations and boosts of the 4+4 spacetime;
- **form Q**, the *quantized-sector form*: the nine invariants are built with the adjoint $\bar{\psi} = \psi^\dagger \beta$ of the positive kinetic term.

The author's instruction of November 13, 2026 for this folder is: *form C is the definition; both options are to be recorded correctly and completely, and explained and discussed.* This document is that record. It contains

1. every definition that is needed to state the two forms (Sections 1 to 3), so that it can be read without any other document;
2. the exact relation between the two forms (Section 4);
3. what each form implies for every calculation of this folder that involves the self-interaction, with the numbers (Sections 5 to 10);
4. a discussion of the two options, with the arguments for and against each (Section 11);
5. what has been established and what has not (Section 12);
6. where each form is in the program and in the data files, and how to repeat every number (Section 13).

The result in one table. For the same nine couplings of the Lagrangian the two forms differ as follows. All entries are derived in the sections named.

property	form C (the definition)	form Q (the alternative)	section
bilinears	$\tilde{\psi}\,\Theta^K\psi$, $\tilde{\psi} = \psi^\dagger C^-$	$\bar{\psi}\,\Theta^K_Q\psi$, $\bar{\psi} = \psi^\dagger\beta$	3
invariance of the quartic term	all 28 rotations and boosts of the 4+4 frame	28 generators of which only 13 are transformations of the 4+4 coordinates	8
two particles of the same half $W = \pm 1$ of the field	same interaction in both forms	same interaction in both forms	4
a particle of each half	opposite sign to form Q	opposite sign to form C	4
an attractive scalar coupling ($g_0 < 0$) acts in uniform matter	on the *difference* of the scalar densities of the halves	on their *sum*	5

property	form C (the definition)	form Q (the alternative)	section
self-bound matter	one half filled (four internal states)	both halves equally filled (eight states)	5
strength at which matter first binds (set A)	2.74 times that of form Q for $D = 4$, 2.51 times for $D = 3$	reference	5
mass that a massless field acquires in vacuum	opposite signs in the two halves (the covariant mass term)	the same sign in both halves	6
contact couplings in the pair along the hidden axis	enter with the opposite sign	enter as on 7+1	7
internal symmetry $SO(4) \times U(1)_E$ of fields in ordinary space	broken to $SO(3)$ by *every* self-interaction	$SO(4)$ kept by every self-interaction, $U(1)_E$ if $g_1^{\text{eff}} = g_7^{\text{eff}} = 0$	8
chiral invariance	if $g_1^{\text{eff}} = g_7^{\text{eff}}$	if $g_0^{\text{eff}} + g_8^{\text{eff}} = 0$	8

Units: $\hbar = c = 1$; where numbers are given, masses, momenta and couplings are in units of the fermion mass m, unless a cutoff Λ is named as the unit. Every number of this document is read by machine from the data files of the program; none is typed by hand.

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1. Conventions

Coordinates. The eight coordinates are $x0, x1, \dots, x7$: $x0$ is the hidden space coordinate, $x1, x2, x3$ are the coordinates of ordinary space, $x4$ is the ordinary time, and $x5, x6, x7$ are the three extra coordinates, which are time-like on the 4+4 spacetime. The flat metric of the 4+4 spacetime is $\eta^{(4+4)} = \text{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$. Indices $A, B = 0, \dots, 7$ refer to an orthonormal frame; $j = 0, 1, 2, 3$ always denotes one of the four space-like and $t = 5, 6, 7$ one of the three extra time-like coordinates.

The deflating spacetime and the clock. In this folder the three extra coordinates deflate: the metric components along them are $g_{tt}^{(4+4)} = -\text{const}_t^2\, e^{-2\,\text{rate}_t\, x4}$, with the three rates $\text{rate}_5 = \text{g55}$, $\text{rate}_6 = \text{g66}$, $\text{rate}_7 = \text{g77}$ (written upright, to distinguish them from metric components). The proper volume of a unit coordinate cell of $x5, x6, x7$ is $V_d(x4) \propto e^{-\text{gsum}\, x4}$ with $\text{gsum} = \text{g55} + \text{g66} + \text{g77}$. For a field that does not depend on $x5, x6, x7$ the field $\widehat{\psi} = \sqrt{V_d}\,\psi$ (the *weighted field*) obeys the equations of flat spacetime with every coupling g_r of the quartic term replaced by $g_r / V_d(x4)$, and with the mass unchanged. All couplings of the self-interaction therefore grow in proportion to e^{τ_d}, where

$$\tau_d = \text{gsum}\,(x4 - x4_{\text{ref}}) \tag{1.1}$$

is the **deflation clock**, counted from a reference moment $x4_{\text{ref}}$ that is named in each use. Statements of this document about a strength of the couplings are therefore statements about a clock reading. From here on ψ denotes the weighted field and g_r the couplings at the moment considered.

The Dirac matrices. The eight matrices $\gamma^A = \gamma^A_{1644}$ of the 4+4 spacetime are real 16×16 matrices with entries 0 and ± 1. In the following displays $+$ stands for $+1$, $-$ for -1 and a dot for 0.

$$\gamma^0 = \begin{pmatrix}
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot \\
+ & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + \\
\cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot
\end{pmatrix} \qquad \gamma^1 = \begin{pmatrix}
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
- & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot
\end{pmatrix}$$

$$\gamma^2 = \begin{pmatrix}
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & - \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & - & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & + & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & + & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & + & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & - & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & - & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & - & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
+ & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot
\end{pmatrix} \qquad \gamma^3 = \begin{pmatrix}
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & - & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
- & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot
\end{pmatrix}$$

$$\gamma^4 = \begin{pmatrix}
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot & \cdot & + \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
+ & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot
\end{pmatrix} \qquad \gamma^5 = \begin{pmatrix}
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & + \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & - & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & - & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & + & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & - & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & + & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & + & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & + & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & + & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
- & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot
\end{pmatrix}$$

$$\gamma^6 = \begin{pmatrix}
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & + \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
- & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot
\end{pmatrix} \qquad \gamma^7 = \begin{pmatrix}
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - & \cdot \\
- & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & - \\
\cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & - & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & + & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot
\end{pmatrix}$$

Different matrices anticommute, and their squares are the signs of the metric:

$$\gamma^A\gamma^B + \gamma^B\gamma^A = 2\,\eta^{(4+4)\,AB}\,1, \qquad (\gamma^j)^2 = +1 \;\; (j = 0, 1, 2, 3), \qquad (\gamma^4)^2 = (\gamma^t)^2 = -1 \;\; (t = 5, 6, 7). \tag{1.2}$$

The matrices $\gamma^0, \ldots, \gamma^3$ are symmetric and $\gamma^4, \ldots, \gamma^7$ are antisymmetric. The following matrices are built from them (all properties are verified by the program in exact integer arithmetic, check `44c-S1`).

matrix	definition	properties
C^-	$\gamma^0\gamma^1\gamma^2\gamma^3$	real, symmetric, $(C^-)^2 = 1$; $C^-\gamma^A(C^-)^{-1} = -(\gamma^A)^T$
β	$-i\,\gamma^4$	Hermitian, $\beta^2 = 1$
α^j	$-\gamma^4\gamma^j$, $j = 0, 1, 2, 3$	Hermitian, square 1; anticommute with one another and with β

matrix	definition	properties
W	$i\,C^{-}\gamma^{4}$	Hermitian, $W^2=1$, trace 0, imaginary and antisymmetric; commutes with $\gamma^0,\dots,\gamma^4$, hence with β and with every α^j; anticommutes with $\gamma^5,\gamma^6,\gamma^7$; $C^{-}=-W\beta$
χ	$\gamma^0\gamma^1\cdots\gamma^7$	the chirality matrix: real, diagonal, $\chi^2=1$; anticommutes with every γ^A

The matrix W is written W_{1644} where it must be distinguished from other objects of the project. It is the central object of this document.

The field. ψ has sixteen complex components, which anticommute with one another (they are Grassmann-valued, as the quantization of a fermion field requires). A product of two components commutes with everything.

Index lists. $K=\{A_1<A_2<\cdots<A_r\}$ denotes a list of r different indices between 0 and 7; $r=|K|$ is its **rank**. There are $1,8,28,56,70,56,28,8,1$ lists of the ranks $0,\dots,8$, together 256.

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2. The quantized sector, and the two halves of the field

2.1 The fields that do not depend on the extra times

The covariant Lagrangian of the free field on the 4+4 spacetime is

$$\mathcal{L}^{(0)}_{\text{cov}}=\tilde{\psi}\,(\gamma^A\partial_A-m)\,\psi,\qquad \tilde{\psi}=\psi^\dagger C^{-}. \tag{2.1}$$

It contains no factor i, because the matrices are real and square to the signs of the metric: applying the operator $\gamma^A\partial_A+m$ to the field equation $(\gamma^A\partial_A-m)\psi=0$ gives $(\eta^{AB}\partial_A\partial_B-m^2)\psi=0$. For a plane wave with the frequency E along $x4$, the momenta p_j along the four space-like coordinates and the wave numbers k_t along the three extra times this is

$$E^2=m^2+\sum_{j=0}^{3}p_j^2-\sum_{t=5}^{7}k_t^2. \tag{2.2}$$

A wave that depends on the extra times lowers E^2, and for $\sum_t k_t^2>m^2+\sum_j p_j^2$ the frequency is imaginary: such a wave grows or decays in the ordinary time. Only the fields that do **not** depend on $x5,x6,x7$ have a real frequency for all momenta and evolve unitarily in $x4$. They form the **quantized sector**, and all statements of this document about particles, matter and the vacuum concern it. (On the deflating spacetime the same holds for the weighted field.)

2.2 The covariant Lagrangian is indefinite

With the matrices of Section 1, $C^{-}\gamma^4=-i\,W$, $C^{-}\gamma^j=-i\,W\alpha^j$ and $C^{-}=-W\beta$. Inserting these into Eq. (2.1) for a field of the quantized sector:

$$\tilde{\psi}\,(\gamma^A\partial_A-m)\,\psi=-\,\psi^\dagger\,W\,(i\,\partial_4-\mathsf{h})\,\psi,\qquad \mathsf{h}=\sum_{j=0}^{3}\alpha^j\,(-i\partial_j)+\beta\,m. \tag{2.3}$$

h is the Hamiltonian of a single particle. The Hermitian matrix W has the eigenvalues $+1$ and -1, eight times each, and it commutes with h. The sixteen components of the field therefore fall into two sets of eight, called here the two **halves** of the field,

$$\psi = \psi_+ + \psi_-, \qquad W\,\psi_\pm = \pm\,\psi_\pm, \tag{2.4}$$

and h acts inside each half. For a momentum $\mathbf{p}$ the Hamiltonian has the eigenvalues $+\sqrt{\mathbf{p}^2+m^2}$ and $-\sqrt{\mathbf{p}^2+m^2}$, each eight times; four of the eight states of positive energy lie in each half. A particle has therefore **eight internal states, four in each half**.

Equation (2.3) says: *the covariant Lagrangian is the ordinary Lagrangian $\psi^\dagger(i\partial_4-\mathsf{h})\psi$ of a Dirac field on the half $W=-1$, and minus that Lagrangian on the half $W=+1$.* The charge density of the covariant Lagrangian is proportional to $\tilde\psi\gamma^4\psi = -i\,\psi^\dagger W\psi$: it counts the particles of one half positively and those of the other negatively.

2.3 The positive Lagrangian

For a free field the sign of a Lagrangian does not change the equation of motion: both halves obey $i\,\partial_4\psi = \mathsf{h}\,\psi$. The quantum theory of the free field is defined by the positive Lagrangian for all sixteen components,

$$\mathcal{L}_0 = \psi^\dagger\left(i\,\partial_4 - \mathsf{h}\right)\psi = \bar\psi\left(i\,\beta\,\partial_4 - \beta\,\mathsf{h}\right)\psi, \qquad \bar\psi = \psi^\dagger\beta, \tag{2.5}$$

in which every state of positive energy has positive norm. Its mass term is $-m\,\bar\psi\psi = -m\,\psi^\dagger\beta\psi$: the same mass for both halves. The mass term of the covariant Lagrangian is different: because $C^- = -W\beta$,

$$-m\,\tilde\psi\psi = +m\,\psi^\dagger\,W\beta\,\psi = m\left(\psi_+^\dagger\beta\psi_+ - \psi_-^\dagger\beta\psi_-\right), \tag{2.6}$$

a mass term of opposite sign in the two halves. For the free field this difference is again invisible, because the spectrum of h does not depend on the sign of m. With a self-interaction the two halves are no longer equivalent, and the quantized theory can be completed in more than one way. That is the subject of this document.

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3. The self-interaction: three ways to write it

3.1 What is common

In every form the self-interaction is a sum of nine terms, one for each rank,

$$\mathcal{L}_{\text{self}} = -\frac{1}{2}\sum_{r=0}^{8} g_r\,I_r, \qquad I_r = \sum_{|K|=r}\left(\psi^\dagger\,\text{Adj}\;\Theta_K\,\psi\right)\left(\psi^\dagger\,\text{Adj}\;\Theta^K\,\psi\right), \tag{3.1}$$

with nine real coupling constants g_r. Here Adj is a fixed matrix (the *adjoint matrix*), Θ^K is a product of the r Dirac matrices of the list K times a phase ζ_r that makes the **bilinear** $\psi^\dagger\,\text{Adj}\,\Theta^K\psi$ real, and $\Theta_K = (\Theta^K)^{-1}$. I_0 is the square of a scalar, I_1 of a vector, I_2 of an antisymmetric tensor, and so on up to I_7 (an axial vector) and I_8 (a pseudoscalar). The forms differ in the adjoint matrix and, with it, in the phases and in the signs.

3.2 Form C, the covariant form (the definition in this folder)

Adjoint. Adj $= C^-$, that is $\tilde{\psi} = \psi^\dagger C^-$.

Matrices.

$$\Theta^K = \zeta_r\, \gamma^{A_1}\gamma^{A_2}\ldots\gamma^{A_r}, \qquad \zeta_r = \begin{cases} i & \text{for } r = 1,2,5,6, \\ 1 & \text{for } r = 0,3,4,7,8, \end{cases} \tag{3.2}$$

$$(\Theta^K)^2 = \varepsilon_K\, 1, \qquad \varepsilon_K = (-1)^r\, \eta_K, \qquad \eta_K = \prod_{A\in K} (\gamma^A)^2, \qquad \Theta_K = (\Theta^K)^{-1} = \varepsilon_K\, \Theta^K. \tag{3.3}$$

The phases are those for which $C^-\Theta^K$ is Hermitian (check `44c-S2`, all 256 lists). The sign ε_K is written sgn_K in some documents of the project.

Invariants and Lagrangian.

$$I^{\mathrm{C}}_r = \sum_{|K|=r} \varepsilon_K \left(\tilde{\psi}\,\Theta^K\psi\right)^2, \qquad \boxed{\mathcal{L}_{\mathrm{C}} = \psi^\dagger \left(i\,\partial_4 - \mathsf{h}\right)\psi - \frac{1}{2}\sum_{r=0}^{8} g_r\, I^{\mathrm{C}}_r} \tag{3.4}$$

The quartic term is the one that is invariant under all 28 rotations and boosts of the 4+4 frame (check `44c-S4`); the kinetic and mass terms are those of the positive Lagrangian (2.5). **This is the definition of the 4+4 model in this folder.**

Field equation. Varying $\psi^\dagger$, with $V_{\mathrm{C}} = \frac{1}{2}\sum_r g_r\, I^{\mathrm{C}}_r$,

$$i\,\partial_4\psi = \mathsf{h}\,\psi + \frac{\partial V_{\mathrm{C}}}{\partial\psi^\dagger} = \mathsf{h}\,\psi + \sum_{r=0}^{8} g_r \sum_{|K|=r} \left(\tilde{\psi}\,\Theta_K\,\psi\right) C^-\Theta^K\psi. \tag{3.5}$$

It conserves the number $\int \psi^\dagger\psi$ of particles of both halves together.

3.3 Form Q, the quantized-sector form (the alternative)

Adjoint. Adj $= \beta$, that is $\bar{\psi} = \psi^\dagger\beta$, the adjoint of the positive Lagrangian (2.5).

Matrices. The bilinears are built from the eight matrices

$$\gamma^A_{44q} = -i\,\gamma^A \quad (A = 0,\ldots,4), \qquad \gamma^A_{44q} = \gamma^A \quad (A = 5,6,7). \tag{3.6}$$

They anticommute with one another, $\gamma^4_{44q} = \beta$, and their squares are $+1$ for $A = 4$ and -1 for the seven others: the squares of the Dirac matrices of a spacetime with one time (check `44q-S1`). In terms of them the positive Lagrangian (2.5) reads $\bar{\psi}\,(i\,\gamma^A_{44q}\partial_A - m)\,\psi$ for fields of the quantized sector. With

$$\Theta^K_Q = \zeta^Q_r\, \gamma^{A_1}_{44q}\ldots\gamma^{A_r}_{44q}, \qquad \zeta^Q_r = \begin{cases} i & \text{for } r = 2,3,6,7, \\ 1 & \text{for } r = 0,1,4,5,8, \end{cases} \qquad (\Theta^K_Q)^2 = \varepsilon^Q_K\, 1, \qquad \varepsilon^Q_K = \prod_{A\in K} (\gamma^A_{44q})^2, \tag{3.7}$$

the matrices $\beta\,\Theta^K_Q$ are Hermitian (check `44q-S2`), and

$$I_r^{\mathrm{Q}} = \sum_{|K|=r} \varepsilon_K^Q \left(\bar\psi\,\Theta_Q^K\psi\right)^2, \qquad \mathcal{L}_{\mathrm{Q}} = \psi^\dagger\left(i\,\partial_4 - \mathsf{h}\right)\psi - \frac{1}{2}\sum_{r=0}^{8} g_r\, I_r^{\mathrm{Q}}. \tag{3.8}$$

Field equation.

$$i\,\partial_4\psi = \mathsf{h}\,\psi + \sum_{r=0}^{8} g_r \sum_{|K|=r} \left(\bar\psi\,\Theta_{Q,K}\,\psi\right)\beta\,\Theta_Q^K\psi, \qquad \Theta_{Q,K} = \varepsilon_K^Q\,\Theta_Q^K. \tag{3.9}$$

This is, term by term, the equation of the model on the spacetime with seven space-like coordinates and one time (the 7+1 spacetime) for fields that do not depend on $x5, x6, x7$: the algebra of the matrices (3.6) is that of the 7+1 Dirac matrices. Every result that is obtained for such fields on 7+1 holds for form Q without change.

3.4 The covariant classical equation (described; not quantized)

If the quartic term of form C is combined with the *covariant* kinetic and mass terms (2.1), the field equation of the quantized sector follows from Eq. (2.3):

$$i\,\partial_4\psi = \mathsf{h}\,\psi - W\,\frac{\partial V_{\mathrm{C}}}{\partial\psi^\dagger}. \tag{3.10}$$

Compared with Eq. (3.5) the interaction term carries the factor $-W$. On the half $W=-1$ the two equations agree. On the half $W=+1$ the covariant equation has the opposite sign of the whole interaction, the same change of sign as in Eq. (2.3). The covariant equation conserves $\psi^\dagger W\psi$ and not $\psi^\dagger\psi$; it is not the equation of a Hamiltonian with states of positive norm when both halves are occupied. It is a consistent classical field equation and is the one that the covariant Lagrangian implies literally, but no quantum theory of particles is built on it in this project.

3.5 The three options side by side

option	kinetic and mass term	bilinears of the quartic term	status in this folder
form C	positive, Eq. (2.5)	$\tilde\psi\,\Theta^K\psi$, $\tilde\psi = \psi^\dagger C^-$, Eq. (3.4)	**the definition**
form Q	positive, Eq. (2.5)	$\bar\psi\,\Theta_Q^K\psi$, $\bar\psi = \psi^\dagger\beta$, Eq. (3.8)	the alternative, computed beside the definition
covariant classical equation	indefinite, Eq. (2.1)	as in form C	described in Section 3.4; not quantized

The hypotheses of the project fix the nine invariants as "the" invariants of the spacetime. On the 7+1 spacetime this determines the model. On the 4+4 spacetime it does not, because the adjoint that makes the invariants covariant (C^-) is not the adjoint that makes the energy positive (β): the two differ by the matrix W.

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4. The exact relation between the two forms

4.1 Form C is form Q with W inserted in every bilinear

For every one of the 256 lists K (check `44c-S11`, exact),

$$C^{-}\Theta^{K} = z_K\, W\, \beta\, \Theta^{K}_{Q}, \qquad z_K \in \{+1, -1, +i, -i\}, \tag{4.1}$$

where $z_K = \pm 1$ if K contains an even number of the indices $5, 6, 7$ (128 lists) and $z_K = \pm i$ if it contains an odd number (128 lists). Consequently

$$I^{\mathrm{C}}_r = \sum_{|K|=r} \varepsilon^{Q}_{K}\, (\psi^{\dagger}\, W\, \beta\, \Theta^{K}_{Q}\, \psi)^2, \qquad \text{to be compared with} \qquad I^{\mathrm{Q}}_r = \sum_{|K|=r} \varepsilon^{Q}_{K}\, (\psi^{\dagger}\, \beta\, \Theta^{K}_{Q}\, \psi)^2. \tag{4.2}$$

The covariant form is the quantized-sector form with the matrix W inserted in every bilinear, and nothing else.

4.2 What the insertion does

Write $M_K = \beta\, \Theta^{K}_{Q}$ (a Hermitian matrix) and decompose the field into its halves (2.4).

Lists with an even number of the indices 5, 6, 7. M_K commutes with W and acts inside each half. With $a = \psi^{\dagger}_{+} M_K \psi_{+}$ and $b = \psi^{\dagger}_{-} M_K \psi_{-}$, form Q contains $(a+b)^2$ and form C contains $(a-b)^2$. The terms a^2 and b^2, which act between two particles of the same half, are equal. The term $2ab$, which acts between a particle of one half and a particle of the other, has opposite signs.

Lists with an odd number. M_K anticommutes with W and connects the two halves. With $c = \psi^{\dagger}_{+} M_K \psi_{-}$, form Q contains $(c + c^{\dagger})^2$ and form C contains $(c - c^{\dagger})^2$. The terms c^2 and $(c^{\dagger})^2$ move two particles from one half to the other; they are equal in the two forms. The term $c\, c^{\dagger} + c^{\dagger} c$ exchanges a particle of one half with a particle of the other; it has opposite signs.

Together:

> **Two particles of the same half interact identically in the two forms. A particle of the half $W = +1$ and a particle of the half $W = -1$ interact with opposite signs.**

Two further consequences of the decomposition are used below. (i) In both forms the interaction can move *two* particles together from one half to the other. The number of particles of each half is therefore not conserved separately; only the total number is, and the difference of the two numbers modulo 2. (ii) Both forms are symmetric under the exchange of the two halves. For fields in ordinary space this exchange is the transformation $\psi \to \gamma^0\gamma^5\psi$: the matrix $\gamma^0\gamma^5$ is Hermitian with square 1, it commutes with $\alpha^1, \alpha^2, \alpha^3$ and β, it anticommutes with W, and it changes every bilinear of either form by a sign at most. For fields that depend on the hidden coordinate too, the same transformation is combined with the reflection $x0 \to -x0$.

4.3 What the two forms have in common

The algebra that follows from the completeness of the 256 matrices is the same in the two forms (checks `44c-S3`, `44c-S5`, `44c-S6` and `44q-S3`, `44q-S5`, `44q-S6`). In particular, because the field components anticommute, the nine invariants obey five linear relations in each form,

$$\begin{aligned} &I_2 = -3I_0 - I_1 - I_7 - I_8, \qquad && I_3 = -4I_0 - I_7 + 4I_8, \qquad && I_4 = -5I_0 - 5I_8, \\ &I_5 = -4I_0 - I_1 + 4I_8, && I_6 = -I_0 + I_1 + I_7 - 3I_8, \end{aligned} \tag{4.3}$$

so that only four invariants are independent, and

$$\mathcal{L}_{\mathrm{self}} = -\tfrac{1}{2}(g^{\mathrm{eff}}_0\, I_0 + g^{\mathrm{eff}}_1\, I_1 + g^{\mathrm{eff}}_7\, I_7 + g^{\mathrm{eff}}_8\, I_8) \tag{4.4}$$

with the four **effective couplings**

$$\begin{aligned} g_0^{\text{eff}} &= g_0 - 3g_2 - 4g_3 - 5g_4 - 4g_5 - g_6, & g_1^{\text{eff}} &= g_1 - g_2 - g_5 + g_6, \\ g_7^{\text{eff}} &= g_7 - g_2 - g_3 + g_6, & g_8^{\text{eff}} &= g_8 - g_2 + 4g_3 - 5g_4 + 4g_5 - 3g_6. \end{aligned} \tag{4.5}$$

The relations (4.3) are consequences of the Fierz rearrangement $\sum_{|K|=r}(a\,\Theta_K b)(c\,\Theta^K d) = \sum_{r'}\mathsf{F}_{rr'}\sum_{|J|=r'}(a\,\Theta_J d)(c\,\Theta^J b)$ with the matrix $\mathsf{F}_{rr'} = \frac{1}{16}\sum_l (-1)^{rr'-l}\binom{r'}{l}\binom{8-r'}{r-l}$, which is the same for both forms: for anticommuting fields $(1+\mathsf{F})\,I = 0$, and F has the eigenvalue $+1$ five times. The combinations (4.5) are the same functions of the nine couplings in both forms. A model is therefore specified, in either form, by the four numbers $g_0^{\text{eff}}, g_1^{\text{eff}}, g_7^{\text{eff}}, g_8^{\text{eff}}$ (scalar, vector, axial-vector and pseudoscalar coupling), and the question "form C or form Q" is the question what these four numbers *mean*.

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5. Uniform matter

5.1 The Hartree–Fock energy

In uniform matter at rest the expectation value of a product of two fields is the density matrix $\mathsf{n}_{ab} = \langle\psi_b^\dagger\psi_a\rangle$. When the two halves are filled with particles up to Fermi momenta k_+ and k_-, and no coherence between the halves is present, it has the form

$$\mathsf{n} = \tfrac{1}{16}(n\,1 + n_S\,\beta + n_W\,W + n_{SW}\,W\beta), \tag{5.1}$$

$$n = n_+ + n_-, \qquad n_W = n_+ - n_-, \qquad n_S = n_{S+} + n_{S-}, \qquad n_{SW} = n_{S+} - n_{S-}, \tag{5.2}$$

where $n_\pm = \langle\psi_\pm^\dagger\psi_\pm\rangle$ are the number densities of the two halves and $n_{S\pm} = \langle\psi_\pm^\dagger\beta\psi_\pm\rangle$ their scalar densities. By Wick's theorem, $\langle(\psi^\dagger M\psi)^2\rangle = [\mathrm{Tr}(M\mathsf{n})]^2 - \mathrm{Tr}(M\mathsf{n}M\mathsf{n})$ (a direct term minus an exchange term), the energy density of the self-interaction is, in both forms, a sum of four squares (checks `44c-S7`, `44q-S7`, exact):

$$\varepsilon_{\text{int}} = \tfrac{1}{2}(T_V\,n^2 + T_S\,n_S^2 + D_V\,n_W^2 + D_S\,n_{SW}^2). \tag{5.3}$$

T_V, T_S are the couplings of the **total** densities, D_V, D_S the couplings of the **differences**. They are four combinations of the couplings,

$$\begin{aligned} G_S^{\text{HF}} &= \tfrac{1}{16}(15\,g_0^{\text{eff}} - 8\,g_1^{\text{eff}} - 8\,g_7^{\text{eff}} - g_8^{\text{eff}}), & G_V^{\text{HF}} &= \tfrac{1}{16}(-g_0^{\text{eff}} + 22\,g_1^{\text{eff}} - 6\,g_7^{\text{eff}} + g_8^{\text{eff}}), \\ G_S^{\text{c}} &= \tfrac{1}{16}(-g_0^{\text{eff}} - 2\,g_1^{\text{eff}} + 2\,g_7^{\text{eff}} + g_8^{\text{eff}}), & G_V^{\text{c}} &= -\tfrac{1}{16}(g_0^{\text{eff}} + g_8^{\text{eff}}), \end{aligned} \tag{5.4}$$

the same four in both forms, with the roles of the totals and of the differences exchanged (check `44-S13`, exact):

coefficient of	form C (the definition)	form Q (the alternative)
n^2	G_V^{c}	G_V^{HF}
n_S^2	G_S^{c}	G_S^{HF}
n_W^2	G_V^{HF}	G_V^{c}
n_{SW}^2	G_S^{HF}	G_S^{c}

In terms of the nine couplings: $G_S^{\rm HF} = \frac{1}{16}(15g_0 - 8g_1 - 28g_2 - 56g_3 - 70g_4 - 56g_5 - 28g_6 - 8g_7 - g_8)$, $G_V^{\rm HF} = \frac{1}{16}(-g_0 + 22g_1 - 14g_2 + 14g_3 - 14g_5 + 14g_6 - 6g_7 + g_8)$, $G_S^{\rm c} = \frac{1}{16}(-g_0 - 2g_1 + 2g_2 + 6g_3 + 10g_5 - 2g_6 + 2g_7 + g_8)$, $G_V^{\rm c} = \frac{1}{16}(-g_0 + 4g_2 + 10g_4 + 4g_6 - g_8)$. $G_S^{\rm HF}$ and $G_V^{\rm HF}$ are the Hartree–Fock couplings of the 7+1 model: the direct term gives g_0 and g_1, and the exchange terms reduce the scalar coupling by one sixteenth and add to the vector coupling. When only g_0, g_1, g_7, g_8 differ from zero, $G_S^{\rm c}$ and $G_V^{\rm c}$ come from exchange terms alone.

An equivalent way to write Eq. (5.3) shows what happens between particles:

$$\varepsilon_{\rm int} = \tfrac{1}{2}\, a_V\, (n_+^2 + n_-^2) + \tfrac{1}{2}\, a_S\, (n_{S+}^2 + n_{S-}^2) \;\pm\; (b_V\, n_+ n_- + b_S\, n_{S+} n_{S-}), \tag{5.5}$$

with the upper sign for form Q and the lower sign for form C, and with

$$\begin{aligned} a_S &= G_S^{\rm HF} + G_S^{\rm c} = \tfrac{1}{8}(7\, g_0^{\rm eff} - 5\, g_1^{\rm eff} - 3\, g_7^{\rm eff}), \\ a_V &= G_V^{\rm HF} + G_V^{\rm c} = \tfrac{1}{8}(-\, g_0^{\rm eff} + 11\, g_1^{\rm eff} - 3\, g_7^{\rm eff}), \\ b_S &= G_S^{\rm HF} - G_S^{\rm c} = \tfrac{1}{8}(8\, g_0^{\rm eff} - 3\, g_1^{\rm eff} - 5\, g_7^{\rm eff} - g_8^{\rm eff}), \\ b_V &= G_V^{\rm HF} - G_V^{\rm c} = \tfrac{1}{8}(11\, g_1^{\rm eff} - 3\, g_7^{\rm eff} + g_8^{\rm eff}). \end{aligned} \tag{5.6}$$

a_S, a_V are the couplings between particles of the same half; they are the same in both forms. b_S, b_V are the couplings between particles of different halves; they enter with opposite signs. This is Section 4.2 again, now for the mean field.

5.2 The two sets of couplings used for the numbers

The numerical results of this document are given for two sets of couplings of the Lagrangian, each with one strength $g > 0$ that grows along the clock:

- **set A**: a scalar attraction alone, $g_0 = -g$, all other couplings zero;
- **set B**: the ratio of the author's example, $g_0 = -g$ and $g_1 = +0.1\, g$, all other couplings zero.

The mean-field couplings are then (in units of g; exact). The last two columns are the couplings between particles of different halves as they stand in the energy: $T_V - D_V$ and $T_S - D_S$, which are $+b_V, +b_S$ in form Q and $-b_V, -b_S$ in form C.

set	form	T_V/g	T_S/g	D_V/g	D_S/g	a_V/g	a_S/g	$(T_V - D_V)/g$	$(T_S - D_S)/g$
A	C	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$-\frac{15}{16}$	$\frac{1}{8}$	$-\frac{7}{8}$	0	1
A	Q	$\frac{1}{16}$	$-\frac{15}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$-\frac{7}{8}$	0	-1
B	C	$\frac{1}{16}$	$\frac{1}{20}$	$\frac{1}{5}$	$-\frac{79}{80}$	$\frac{21}{80}$	$-\frac{15}{16}$	$-\frac{11}{80}$	$\frac{83}{80}$
B	Q	$\frac{1}{5}$	$-\frac{79}{80}$	$\frac{1}{16}$	$\frac{1}{20}$	$\frac{21}{80}$	$-\frac{15}{16}$	$\frac{11}{80}$	$-\frac{83}{80}$

In form Q the attraction of the scalar coupling stands in T_S: it lowers the energy of *any* scalar density. In form C it stands in D_S: it lowers the energy only if the scalar densities of the two halves *differ*; the total scalar density is weakly repelled ($T_S = g/16 > 0$ for set A).

5.3 The Kohn–Sham equations of two halves

In the mean field (5.3) the particles of the half $W = \pm 1$ move with a mass $M_\pm$ (the Kohn–Sham mass) and an energy shift $V_\pm$,

$$M_\pm = m + T_S\, n_S \pm D_S\, n_{SW}, \qquad V_\pm = T_V\, n \pm D_V\, n_W, \tag{5.7}$$

and the densities are those of free particles of these masses filled up to the Fermi momenta $k_\pm$. In D space dimensions, with $w_\pm(k) = \sqrt{k^2 + M_\pm^2}$ and four internal states per half,

$$n_\pm = 4\int_{|\mathbf{k}|<k_\pm} \frac{d^D k}{(2\pi)^D}, \qquad n_{S\pm} = 4\int_{|\mathbf{k}|<k_\pm} \frac{d^D k}{(2\pi)^D}\,\frac{M_\pm}{w_\pm}, \tag{5.8}$$

$$\varepsilon = \sum_\pm 4\int_{|\mathbf{k}|<k_\pm} \frac{d^D k}{(2\pi)^D}\, w_\pm + \tfrac{1}{2}(T_V\, n^2 + D_V\, n_W^2) - \tfrac{1}{2}(T_S\, n_S^2 + D_S\, n_{SW}^2), \tag{5.9}$$

$$p = \sum_\pm 4\int_{|\mathbf{k}|<k_\pm} \frac{d^D k}{(2\pi)^D}\,\frac{k^2}{D\, w_\pm} + \Pi, \qquad \Pi = \tfrac{1}{2}(T_V\, n^2 + D_V\, n_W^2) + \tfrac{1}{2}(T_S\, n_S^2 + D_S\, n_{SW}^2), \tag{5.10}$$

$$\mu_\pm = \sqrt{k_\pm^2 + M_\pm^2} + V_\pm. \tag{5.11}$$

Here ε is the energy density, p the pressure along the coordinates in which the particles move, Π the **interaction pressure** (it is also the pressure along the deflating coordinates, along which the particles of the quantized sector have no momentum), and $\mu_\pm$ the chemical potentials of the two halves. The two equations (5.7) for M_+ and M_-, with the scalar densities (5.8), are the **gap equations**. $D = 4$ is the quantized sector itself (momenta along $x0, x1, x2, x3$); $D = 3$ describes fields that do not depend on the hidden coordinate $x0$ either.

The polarization. Because the interaction can move two particles from one half to the other (Section 4.2), the numbers of the two halves are not fixed separately. At a given total density n the state is the one of lowest energy with respect to the **polarization**

$$\mathrm{pol}_W = \frac{n_+ - n_-}{n} \in [0, 1] \tag{5.12}$$

(the sign is free by the symmetry between the halves). If both halves are occupied the minimum has $\mu_+ = \mu_-$.

The approximation. These equations fill the states of positive energy only (the **no-sea approximation**); the filled states of negative energy are treated in Section 6. Among the solutions of the gap equations the program takes those with non-negative masses, which are the continuation of the dilute gas. Such a solution is a regular function of the density and of the couplings as long as the eigenvalues of the **response matrix** of the gap equations,

$$R = \begin{pmatrix} a_S\,\partial n_{S+}/\partial M_+ & b_S\,\partial n_{S-}/\partial M_- \\ b_S\,\partial n_{S+}/\partial M_+ & a_S\,\partial n_{S-}/\partial M_- \end{pmatrix}, \tag{5.13}$$

stay below 1 (the derivatives at fixed Fermi momenta; for form C the off-diagonal elements carry the opposite sign, which does not change the eigenvalues). For equal halves the eigenvalues are $2\,T_S\,\partial n_{S+}/\partial M$ and $2\,D_S\,\partial n_{S+}/\partial M$; for one filled half they are $a_S\,\partial n_{S+}/\partial M$ and 0.

5.4 Form C polarizes matter; form Q does not

For the same couplings of the Lagrangian, the energy of matter of a given density was computed as a function of the polarization. The table gives the energy per particle minus the mass, $\varepsilon/n - m$ in units of m, at the total density of the Fermi momentum $k_{F,\mathrm{tot}} = 0.6\,m$ (the Fermi momentum that the density would have with equal halves), for strengths 1.5 and 3 times the strength at which form Q first binds matter:

D	set	strength	C, $\mathrm{pol}_W = 0$	C, 0.5	C, 1	Q, $\mathrm{pol}_W = 0$	Q, 0.5	Q, 1
4	A	1.5	+0.1358	+0.1155	+0.0598	−0.0159	+0.0016	+0.0598
4	A	3	+0.1588	+0.1121	−0.0099	−0.1137	−0.0897	−0.0099
4	B	1.5	+0.1436	+0.1187	+0.0543	−0.0343	−0.0138	+0.0543
4	B	3	+0.1745	+0.1239	+0.0179	−0.1044	−0.0764	+0.0179
3	A	1.5	+0.1264	+0.1074	+0.0553	−0.0380	−0.0158	+0.0553
3	A	3	+0.1511	+0.1039	−0.0167	−0.1430	−0.1132	−0.0167
3	B	1.5	+0.1301	+0.1107	+0.0612	−0.0406	−0.0164	+0.0612
3	B	3	+0.1583	+0.1140	+0.0197	−0.1188	−0.0864	+0.0197

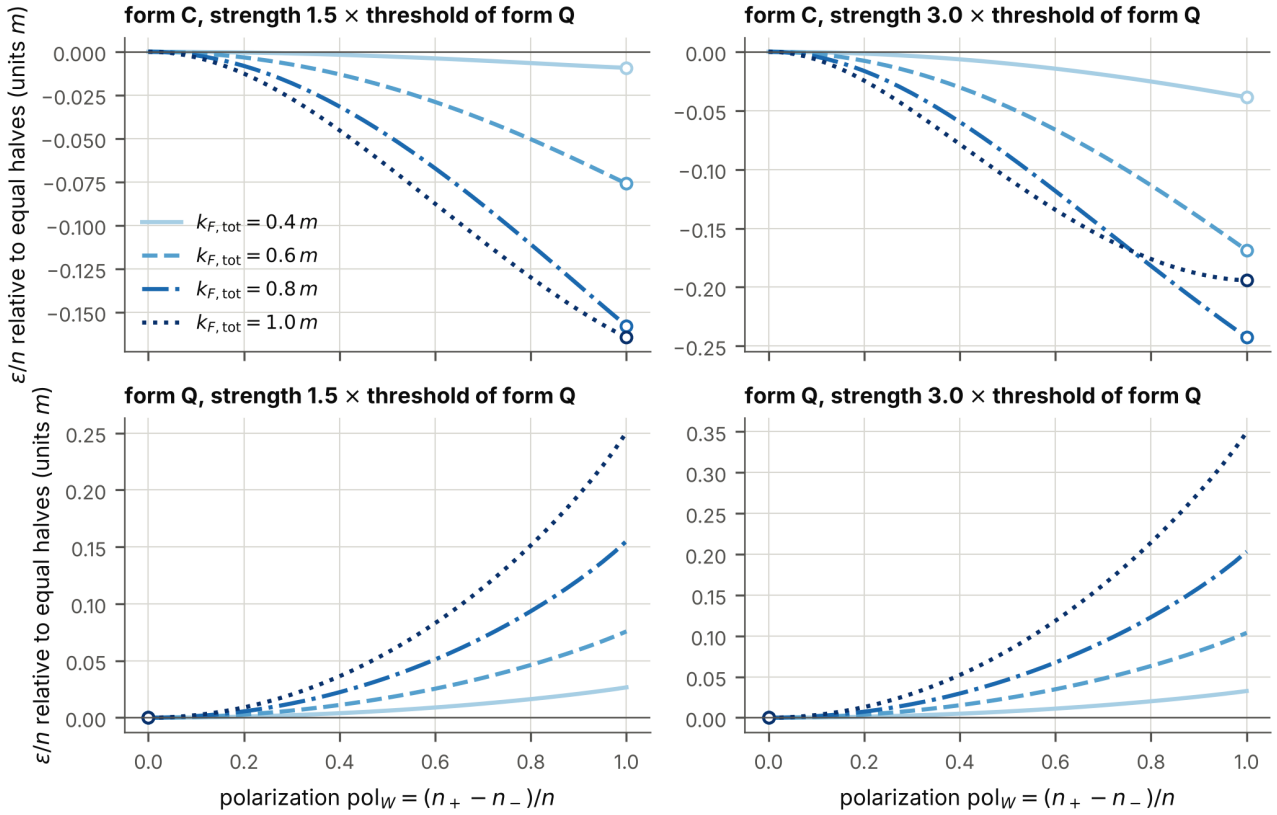


Figure 1: Energy per particle of matter of a given total density against the polarization of the two halves, relative to its value for equal halves ($D = 4$, set A). Upper row: form C; lower row: form Q; left: 1.5 times, right: 3 times the strength at which form Q first binds matter. The four curves are four total densities (light to dark and by line style); the circle marks the lowest point of each curve. In form C the lowest state has one half filled, or nearly so at the highest density; in form Q it has equal halves.

In form C the energy falls with the polarization; in form Q it rises (Figure 1). Over 48 cases (two dimensions, two sets, three strengths, four densities) the state of lowest energy is polarized in 48 cases in form C and in 0 cases in form Q (check `fm-F3`). The reason is Eq. (5.5). For an attractive scalar coupling $a_S < 0$ and $b_S < 0$. In form C a particle is attracted by the particles of its own half (coupling a_S) and repelled by those of the other half (coupling $-b_S > 0$), so that the particles gather in one half. In form Q a particle is attracted by both.

A limit of the calculation in form Q. Of the 3936 states examined, 108 states of form Q with unequal halves do not exist: at 3 and 6 times the threshold and at high density the attraction by the more

populated half, b_S, which is stronger than the attraction a_S inside a half, would make the mass of the particles of the less populated half negative, and the gap equations have no solution with non-negative masses. The states with equal halves and with one filled half exist in all cases. In form C every state exists.

5.5 The strength at which matter first binds

Matter is **self-bound** when its energy per particle, minimized over the density and the polarization, lies below the mass m. The smallest strength g at which this happens is the binding threshold; on the deflating spacetime it is reached at a definite clock reading.

D	set	form	$g\,m^{D-1}$	state	$k_{F,\mathrm{tot}}/m$	M_+/m	M_-/m	$\Lambda_{\max}/m$
4	A	C	224.7386	one half	0.6140	0.5239	1.5441	0.520
4	A	Q	82.1116	equal halves	0.8054	0.4946	0.4946	0.727
4	B	C	332.2917	one half	0.5211	0.5625	1.4842	0.449
4	B	Q	121.0858	equal halves	0.6792	0.5424	0.5424	0.628
3	A	C	24.5717	one half	0.5374	0.6454	1.4053	0.463
3	A	Q	9.8090	equal halves	0.7356	0.6276	0.6276	0.733
3	B	C	31.1674	one half	0.4635	0.6712	1.3638	0.400
3	B	Q	12.3819	equal halves	0.6351	0.6575	0.6575	0.635

($g\,m^{D-1}$ is the dimensionless strength; "state" is the state of lowest energy just above the threshold; $M_\pm$ are the masses of the particles of the two halves, for an empty half the mass that a particle of that half would have; $\Lambda_{\max}$ is explained in Section 6.) In form C the bound state has one half filled, in form Q equal halves, in all eight cases (check `fm-F2`). Each threshold is verified directly (bound 0.1 per cent above, not bound 0.1 per cent below) and agrees with the threshold of matter of one kind of particle computed by other routines, to 5.7×10^{-14}: matter of form C with one filled half is matter of *one kind of particle with four internal states and the couplings* a_S, a_V; matter of form Q with equal halves is matter of one kind with eight states and the couplings $T_S = G_S^{\mathrm{HF}}$, $T_V = G_V^{\mathrm{HF}}$.

The ratio of the threshold of form C to that of form Q is 2.74 for set A and 2.74 for set B in $D=4$, and 2.51 and 2.52 in $D=3$. On the clock, form C binds matter later by $\Delta\tau_d = 1.01$ (set A) and 1.01 (set B) in $D=4$, and by 0.92 and 0.92 in $D=3$. Two things make the difference. Only four of the eight internal states take part. And the attraction inside one half, $a_S = \frac{7}{8}\,g_0$ for set A, is weaker than the attraction between the halves, $b_S = g_0$, from which no exchange term is subtracted; matter of form Q with equal halves has the benefit of both, $T_S = \frac{1}{2}(a_S + b_S)$.

5.6 Self-bound matter along the clock

With the clock counted from the binding threshold of each case, so that $g = g_{\mathrm{threshold}}\,e^{\tau_d}$, the saturated state (the state of lowest energy per particle) develops as follows. The table gives $\varepsilon/n - m$ in units of m at four clock readings, and, at the clock readings 1 and 3, the mass M of the particles of the filled half or halves and the Fermi momentum $k = k_{F,\mathrm{tot}}$ that the density would have with equal halves (both in units of m).

D	set	form	$\tau_d=1$	2	3	4	M at 1	M at 3	k at 1	k at 3
4	A	C	−0.1092	−0.2079	−0.2913	−0.3597	0.351	0.183	0.538	0.361
4	A	Q	−0.1284	−0.2449	−0.3438	−0.4254	0.319	0.156	0.713	0.486
4	B	C	−0.0829	−0.1577	−0.2209	−0.2727	0.392	0.215	0.452	0.301
4	B	Q	−0.0968	−0.1842	−0.2580	−0.3185	0.371	0.199	0.592	0.395
3	A	C	−0.1226	−0.2400	−0.3365	−0.4116	0.373	0.157	0.502	0.308

D	set	form	$\tau_d = 1$	2	3	4	M at 1	M at 3	k at 1	k at 3
3	A	Q	−0.1424	−0.2801	−0.3945	−0.4844	0.347	0.137	0.699	0.442
3	B	C	−0.0940	−0.1835	−0.2567	−0.3133	0.407	0.183	0.426	0.256
3	B	Q	−0.1093	−0.2136	−0.2990	−0.3652	0.389	0.170	0.588	0.357

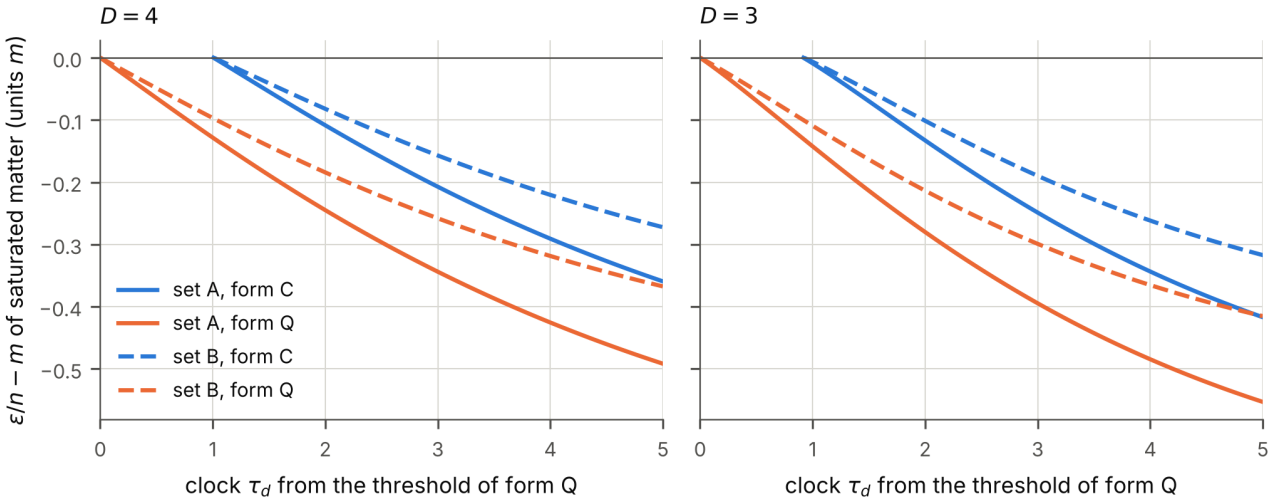


Figure 2: Energy per particle minus the mass of self-bound (saturated) matter along the deflation clock, for the same couplings of the Lagrangian in both forms. The clock is counted here from the moment at which form Q first binds matter of the same set, so that the curves of form C begin later, where form C first binds. Solid lines: set A; dashed lines: set B; blue: form C (one half filled); orange: form Q (equal halves). Left: $D = 4$; right: $D = 3$.

Figure 2 shows the same development on one clock for both forms. At all 101 clock readings from 0 to 5 the kind of the state stays the same (one half filled in form C, equal halves in form Q), the binding energy grows monotonically, and the pressure vanishes (check `fm-F4`). Counted from its own threshold, matter of form C is somewhat less strongly bound than matter of form Q at the same clock reading. The largest eigenvalue of the response matrix (5.13) along the table is 0.881: every state is the regular continuation of the dilute gas. Beyond the table the saturated state of form Q reaches the eigenvalue 1, at which its two masses can split, at these clock readings: $D = 4$, set A: $\tau_d = 6.416$; $D = 4$, set B: not before $\tau_d = 9$; $D = 3$, set A: $\tau_d = 5.296$; $D = 3$, set B: $\tau_d = 7.323$. The saturated state of form C has a negative eigenvalue throughout.

5.7 Matter of fixed density while the couplings grow

A gas of a fixed number of particles per coordinate volume has a fixed density of the weighted field while the couplings grow. The table gives the clock readings, counted from the binding threshold of each case, at which the two halves become unequal, at which one half is empty, and at which the pressure p becomes negative (the matter is then under tension and unstable against clumping). A dash means that the event does not happen for $-3 \le \tau_d \le 5$.

D	set	form	$k_{F,\mathrm{tot}}/m$	halves unequal from	one half empty from	pressure negative from
4	A	C	0.3	−0.53	−0.17	+0.31
4	A	C	0.6	−1.77	−1.43	−0.43
4	A	Q	0.3	—	—	+0.78
4	A	Q	0.6	—	—	−0.36

D	set	form	$k_{F,\mathrm{tot}}/m$	halves unequal from	one half empty from	pressure negative from
4	B	C	0.3	-0.82	-0.47	$+0.06$
4	B	C	0.6	-1.98	-1.62	—
4	B	Q	0.3	—	—	$+0.51$
4	B	Q	0.6	—	—	-0.49
3	A	C	0.3	-0.93	-0.72	-0.03
3	A	C	0.6	-1.48	-1.27	—
3	A	Q	0.3	—	—	$+0.21$
3	A	Q	0.6	—	—	-0.21
3	B	C	0.3	-1.02	-0.88	-0.12
3	B	C	0.6	-1.52	-1.32	—
3	B	Q	0.3	—	—	$+0.10$
3	B	Q	0.6	—	—	-0.14

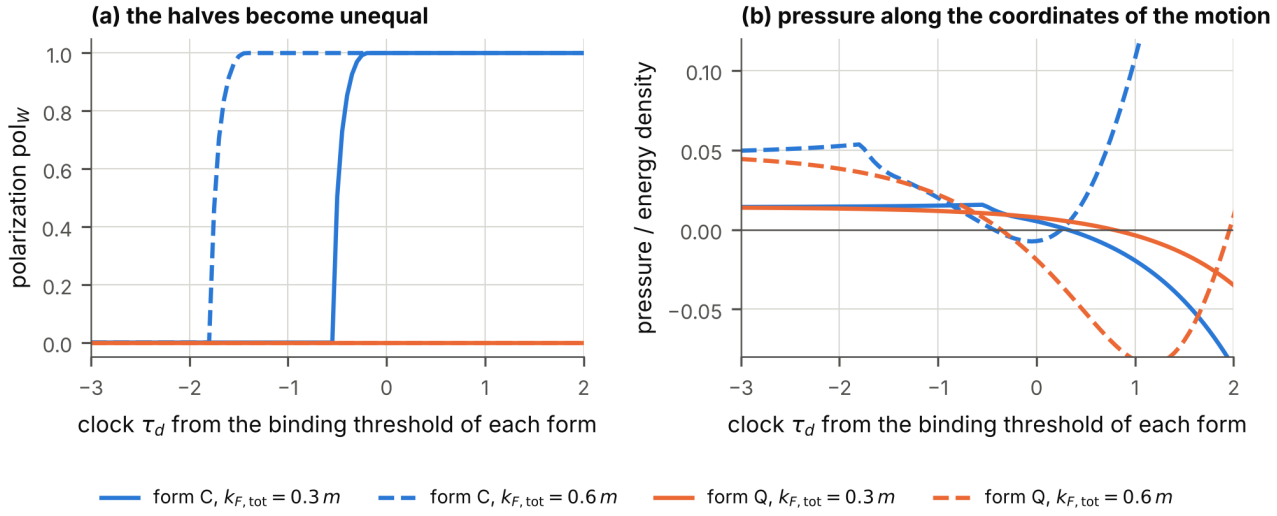


Figure 3: Matter of a fixed total density while the couplings grow ($D = 4$, set A; the clock of each form is counted from its own binding threshold). (a) The polarization: in form C the two halves become unequal at a definite clock reading and one half empties soon after; in form Q they stay equal. (b) The pressure along the coordinates of the motion, divided by the energy density. Solid lines: $k_{F,\mathrm{tot}} = 0.3\,m$; dashed lines: $0.6\,m$.

In form C the gas polarizes before the binding threshold is reached (Figure 3), within at most 0.35 of the clock from the first inequality of the halves to an empty half. Where its pressure becomes negative, that happens after the polarization; in 3 of the 8 cases (the higher density) the pressure stays positive, because the repulsion a_V between the particles of the filled half grows with the couplings while the attraction saturates as the mass falls. In form Q the halves stay equal and the pressure becomes negative in all cases (check `fm-F5`).

5.8 The effective potential of Cornwall, Jackiw and Tomboulis for two halves

The same states follow from an effective potential (the CJT method in the Hartree–Fock truncation). For two halves it is a function of the two masses and the two energy shifts at a given chemical potential μ,

$$\Omega(M_+,M_-,V_+,V_-;\mu)=\sum_{\pm}\Omega_1(M_\pm,\mu-V_\pm)-\frac{\mu_S^2}{2\,T_S}-\frac{d_S^2}{2\,D_S}-\frac{v_T^2}{2\,T_V}-\frac{v_D^2}{2\,D_V}, \tag{5.14}$$

$$\mu_S=\tfrac{1}{2}(M_++M_-)-m,\quad d_S=\tfrac{1}{2}(M_+-M_-),\quad v_T=\tfrac{1}{2}(V_++V_-),\quad v_D=\tfrac{1}{2}(V_+-V_-), \tag{5.15}$$

where $\Omega_1(M,\mu')=-4\int_{w<\mu'}d^Dk/(2\pi)^D\,(\mu'-w)$ is the grand potential of free particles of the mass M with four internal states. Its four stationarity conditions are the equations (5.7), and at the stationary point $-\Omega$ is the pressure and $\Omega+\mu\,n$ the energy density. For 80 states of both forms (40 with equal halves, 17 with unequal halves, 23 with one half filled) the stationary point of Ω, found by Newton's method from displaced start values, agrees with the Kohn–Sham state in both masses, density, energy density and pressure to 4.9×10^{-11} (check `fm-F6`). The two methods are equivalent in both forms.

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6. The vacuum with the filled states of negative energy

6.1 The gap equations of the vacuum

When the states of negative energy are filled up to a cutoff Λ of the momentum, the vacuum itself has scalar densities. The gap equations (5.7) become

$$\tfrac{1}{2}(M_++M_-)-m=T_S\,(s_++s_-),\qquad \tfrac{1}{2}(M_+-M_-)=D_S\,(s_+-s_-),\qquad s_\pm=-4\int_{|\mathbf{k}|<\Lambda}\frac{d^Dk}{(2\pi)^D}\,\frac{M_\pm}{w_\pm}. \tag{6.1}$$

The scalar density of the filled negative-energy states has the sign opposite to the mass. For a field without mass term ($m=0$) the solution $M_+=M_-=0$ always exists. A second solution appears when the more attractive of the two scalar couplings exceeds the **critical coupling**

$$G_c=-\frac{(2\pi)^D\,(D-1)}{8\,A_D\,\Lambda^{D-1}}, \tag{6.2}$$

with A_D the area of the unit sphere in D dimensions ($A_4=2\pi^2$, $A_3=4\pi$): $G_c\Lambda^3=-3\pi^2$ for $D=4$ and $G_c\Lambda^2=-\pi^2/2$ for $D=3$.

6.2 A massless field acquires the mass term of its form

For the sets A and B the attractive scalar coupling is $G_S^{\rm HF}<0$ in both forms, and the critical strength $g_{\rm crit}=G_c/(G_S^{\rm HF}/g)$ is the same for both. The table gives it, and the masses of the two halves 2 per cent above it, in units of the cutoff.

D	set	$G_c\Lambda^{D-1}$	$g_{\rm crit}\Lambda^{D-1}$	form Q: $(M_+,M_-)/\Lambda$	form C: $(M_+,M_-)/\Lambda$
4	A	-29.6088	31.5827	(0.1243, 0.1243)	(0.1243, -0.1243)
4	B	-29.6088	29.9836	(0.1243, 0.1243)	(0.1243, -0.1243)
3	A	-4.9348	5.2638	(0.0860, 0.0860)	(0.0860, -0.0860)
3	B	-4.9348	4.9973	(0.0860, 0.0860)	(0.0860, -0.0860)

Below the critical strength the field stays massless in both forms. Above it,

in form Q the two halves acquire the same mass, $M_+ = M_-$: the mass term $-M\,\bar{\psi}\psi$ of the positive Lagrangian. In form C they acquire masses of opposite sign, $M_+ = -M_-$: by Eq. (2.6) this is the mass term $-M\,\tilde{\psi}\psi$ of the covariant Lagrangian.

(Check `fm-F7`. The solutions of lowest Ω are degenerate by the exchange of the halves and, without mass term, by the reversal of both signs; the table lists the representative with $M_+ > 0$.) In form C the attraction stands in D_S, which couples to the *difference* of the scalar densities; a difference is what a mass of opposite sign in the two halves produces. Each form generates dynamically the mass term that belongs to its own adjoint. The magnitude of the mass is the same in the two forms.

6.3 With a bare mass: a crossover in form Q, a transition in form C

The bare mass term of both forms is $-m\,\bar{\psi}\psi$ (the positive Lagrangian), the same for both halves. Along the clock, counted from the critical strength of the massless field, the masses in vacuum develop as follows (set A; in units of Λ; a single number means $M_+ = M_-$, a pair is (M_+, M_-)):

D	m_{bare}/Λ	form	$\tau_d = -0.5$	0.1	0.5	1
4	0	Q	0	0.311	0.944	1.871
4	0	C	0	(0.311, −0.311)	(0.944, −0.944)	(1.871, −1.871)
4	0.05	Q	0.123	0.492	1.025	1.930
4	0.05	C	0.048	(0.350, −0.255)	(0.992, −0.894)	(1.920, −1.821)
3	0	Q	0	0.241	0.809	1.643
3	0	C	0	(0.241, −0.241)	(0.809, −0.809)	(1.643, −1.643)
3	0.05	Q	0.121	0.426	0.892	1.703
3	0.05	C	0.048	(0.278, −0.183)	(0.855, −0.758)	(1.692, −1.593)

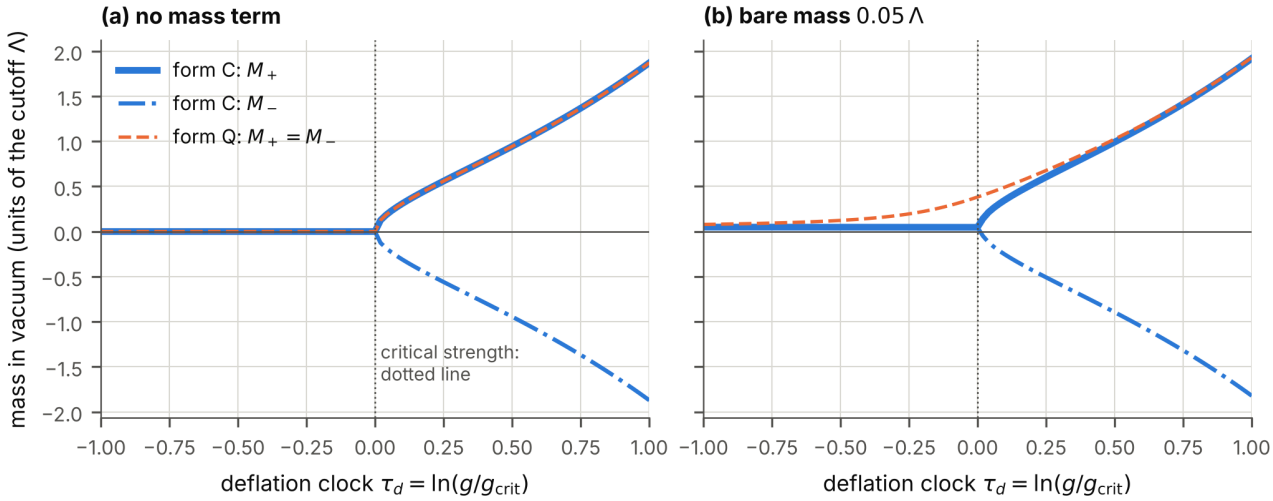


Figure 4: The masses of the particles of the two halves in vacuum with the filled states of negative energy, along the clock counted from the critical strength of a massless field (dotted line; $D = 4$, set A; units of the cutoff). (a) A field without mass term: above the critical strength form Q gives both halves the same mass, form C gives them masses of opposite sign. (b) With the bare mass $0.05\,\Lambda$: form Q passes smoothly through the critical strength; in form C the two masses are equal up to a sharp transition and separate there.

In form Q the bare mass and the dynamical mass have the same structure, and the mass grows smoothly through the critical strength. In form C they have different structures. Below a definite

strength the two masses are equal and slightly smaller than the bare mass ($T_S > 0$ repels the total scalar density). At that strength the masses separate: there is a sharp transition, at which the exchange symmetry of the two halves is broken spontaneously. With the bare mass $0.05\,\Lambda$ it lies at $D = 4$, set A: $\tau_d = 0.009$; $D = 4$, set B: $\tau_d = 0.009$; $D = 3$, set A: $\tau_d = 0.019$; $D = 3$, set B: $\tau_d = 0.020$, and it agrees with the linear criterion $1 + 2\,D_S \cdot 4 \int^{\Lambda} d^D k/(2\pi)^D \; k^2/w^3 = 0$ at the common mass to 2.1×10^{-10} (check `fm-F7`).

6.4 Where the no-sea approximation of Section 5 applies

The matter of Section 5 was computed without the filled negative-energy states. That is consistent only while the vacuum is stable, that is while the most attractive scalar coupling stays below $|G_c|$ of Eq. (6.2) for the cutoff of the model. The column Λ_{max} of the table of thresholds gives, for each case, the largest cutoff (in units of m) for which this holds *at the binding threshold*. It is below m in all eight cases, and smaller in form C than in form Q, because form C needs stronger couplings to bind matter: *in both forms, and more so in form C, self-bound matter by contact couplings alone lies outside the range in which the no-sea approximation is controlled*. This limitation is the same as on the 7+1 spacetime; form C does not remove it.

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7. Pairs

7.1 The two kinds of vector pairs

The superfluid of the project is a condensate of pairs. A pair of two particles with the momenta **k** and $-\mathbf{k}$ is created by an operator $\psi^T X \psi$ with an antisymmetric matrix X. The seven **vector pairs** have $X = C^- \gamma^A$ with $A \neq 4$: one along each of the seven axes other than the ordinary time. The pair along the hidden axis, $X = C^- \gamma^0$, is the spin-singlet pair that the project studies; it and the three pairs along $x5, x6, x7$ are scalars under the rotations of ordinary space, and the pairs along $x1, x2, x3$ form a vector of ordinary space.

The pair density $\kappa_{ab} = \langle \psi_a \psi_b \rangle$ of a vector pair has a definite behavior under W (check `44c-S9`):

$$\begin{aligned} W\,\kappa\,W^T &= -\kappa \quad \text{for the pairs along } x0, x1, x2, x3, \\ W\,\kappa\,W^T &= +\kappa \quad \text{for the pairs along } x5, x6, x7. \end{aligned} \tag{7.1}$$

A pair along one of the four space-like axes consists of **one particle of each half**; a pair along one of the three extra time-like axes consists of **two particles of the same half**. By Section 4.2 the two forms therefore agree on the pairs along $x5, x6, x7$ and have opposite signs on the pairs along $x0, \ldots, x3$.

7.2 Two particles at rest

Two particles at rest, with eight internal states each, form $8 \cdot 7/2 = 28$ antisymmetric states. Each is an eigenstate of the interaction, with the eigenvalue G_{pair} (the interaction energy of the pair times the volume; attractive if negative). In exact arithmetic (checks `44c-S9`, `44q-S9`):

form	number of pairs	the two particles are in	G_{pair}
C	4	different halves	$-g_0^{\text{eff}} - g_1^{\text{eff}} - 5\,g_7^{\text{eff}}$
C	12	different halves	$-g_0^{\text{eff}} - g_1^{\text{eff}} + 3\,g_7^{\text{eff}}$
C	9	same half	$g_0^{\text{eff}} + g_1^{\text{eff}} - 3\,g_7^{\text{eff}}$
C	3	same half	$g_0^{\text{eff}} + g_1^{\text{eff}} + 5\,g_7^{\text{eff}}$

form	number of pairs	the two particles are in	G_{pair}
Q	12	different halves	$g_0^{\text{eff}} + g_1^{\text{eff}} - 3\,g_7^{\text{eff}}$
Q	4	different halves	$g_0^{\text{eff}} + g_1^{\text{eff}} + 5\,g_7^{\text{eff}}$
Q	9	same half	$g_0^{\text{eff}} + g_1^{\text{eff}} - 3\,g_7^{\text{eff}}$
Q	3	same half	$g_0^{\text{eff}} + g_1^{\text{eff}} + 5\,g_7^{\text{eff}}$

In form Q the seven vector pairs are degenerate, with $G_{\text{pair}} = g_0^{\text{eff}} + g_1^{\text{eff}} + 5\,g_7^{\text{eff}}$. In form C they split into the four pairs along the space-like axes, with $-(g_0^{\text{eff}} + g_1^{\text{eff}} + 5\,g_7^{\text{eff}})$, and the three pairs along the extra axes, with $+(g_0^{\text{eff}} + g_1^{\text{eff}} + 5\,g_7^{\text{eff}})$. *A scalar coupling that attracts in form Q repels in the hidden-axis pair of form C.*

7.3 Pairs with momentum: the amplitudes

For particles of mass M with the momenta $\mathbf{p}, -\mathbf{p} \to \mathbf{p}', -\mathbf{p}'$ in ordinary space and the energies E, E', the amplitudes of the four independent contact channels in the pair along the hidden axis and in the pairs along $x5, x6, x7$ (the four pairs whose axis is perpendicular to ordinary space) are, in form Q and on 7+1,

$$\begin{aligned} F_0 &= \frac{EE' + M^2 - \mathbf{p}\cdot\mathbf{p}'}{2\,EE'}, & F_8 &= -\frac{EE' - M^2 - \mathbf{p}\cdot\mathbf{p}'}{2\,EE'}, \\ F_1 &= \frac{3\,EE' - 2M^2 - \mathbf{p}\cdot\mathbf{p}'}{EE'}, & F_7 &= \frac{3\,EE' + 2M^2 - \mathbf{p}\cdot\mathbf{p}'}{EE'}, \end{aligned} \tag{7.2}$$

and the pairing potential of the contact interaction is

$$V_{\text{contact}} = s\,(g_0^{\text{eff}} F_0 + g_1^{\text{eff}} F_1 + g_7^{\text{eff}} F_7 + g_8^{\text{eff}} F_8), \tag{7.3}$$

attractive where negative, with the sign

$$\begin{aligned} s = +1 &: \quad \text{form Q, all four pairs; form C, the pairs along } x5, x6, x7; \\ s = -1 &: \quad \text{form C, the pair along the hidden axis } x0. \end{aligned} \tag{7.4}$$

This holds at all momenta, verified by direct evaluation with the 256 matrices of each form and the projectors on the states of positive energy (check `pc-P5`), and independently with projectors built from numerical eigenvectors (check `Icks-F3`). At rest $(F_0, F_1, F_7, F_8) = (1, 1, 5, 0)$, which is Section 7.2.

7.4 The exchange of the Vorton-Scalar is not affected

The complex scalar field of the project (the Vorton-Scalar, with the coupling g_{VS}) couples to the fermions through $\bar\psi\psi$ and $\bar\psi\, i\chi\, \psi$, with the adjoint $\bar\psi = \psi^\dagger\beta$ in both forms (Section 9). The exchange of its radial mode (mass m_s) and of its phase mode (mass m_ϑ) gives the pairing potential

$$V_{VS} = -g_{VS}^2 \left[\frac{F_S}{q^2 + m_s^2} + \frac{F_P}{q^2 + m_\vartheta^2}\right], \qquad F_S = F_0, \quad F_P = -F_8, \quad \mathbf{q} = \mathbf{p}' - \mathbf{p}, \tag{7.5}$$

in each of these four pairs and in both forms: the two halves enter $\bar\psi\psi$ alike. *In form C the contact self-interaction therefore distinguishes the pair along the hidden axis from the pairs along $x5, x6, x7$, and the exchange does not.*

7.5 The gap

The gap Δ at the Fermi surface follows from the gap equation

$$\Delta(k) = \frac{1}{4\pi^2}\int_0^K dk'\, k'^2\, \mathsf{W}(k,k')\,\frac{\Delta(k')}{\sqrt{\xi(k')^2+\Delta(k')^2}}, \qquad \mathsf{W}(k,k') = \frac{1}{2}\int_{-1}^{1} d\cos\theta\,\big[-V\big], \tag{7.6}$$

with $\xi = \sqrt{k^2+M^2} - \sqrt{k_F^2+M^2}$, the potential $V = V_{VS} + V_{\text{contact}}$, the angle θ between $\mathbf{k}$ and $\mathbf{k}'$, and an upper limit K of the momentum integral (Section 7.6). Each contact coupling is given the range $1/m_{\text{ct}}$ of an exchanged heavy particle, $g \to g\, m_{\text{ct}}^2/(q^2+m_{\text{ct}}^2)$, with $m_{\text{ct}} = m$ in the numbers below. All couplings grow as e^{τ_d}: $g_{VS} = e^{\tau_d/2}$, and the contact couplings are given in units of g_{VS}^2/m^2. The table gives the gap in units of the Fermi kinetic energy at the clock reading 2.04, for $k_F = 0.5\,m$ and two masses of the radial mode, for the two signs s of Eq. (7.4), together with the ultraviolet strength of Section 7.6 (for $m_{_S} = m$).

interaction	$m_{_S} = 0.3m$, $s=+1$	$s=-1$	$m_{_S} = m$, $s=+1$	$s=-1$	$U/4\pi^2$, $s=+1$	$s=-1$
exchange of the Vorton-Scalar alone	0.4118	0.4118	0.0247	0.0247	0.285	0.285
with $g_0^{\text{eff}} = -g_{VS}^2/m^2$	0.9035	0.1481	0.2975	0.0000	0.428	0.143
with $g_0^{\text{eff}} = -g_{VS}^2/m^2$, $g_1^{\text{eff}} = +0.1\, g_{VS}^2/m^2$	0.7407	0.1875	0.1963	0.0000	0.304	0.267
with $g_1^{\text{eff}} = +0.3\, g_{VS}^2/m^2$	0.2370	0.8481	0.0006	0.2285	-0.088	0.659
with $g_7^{\text{eff}} = -0.1\, g_{VS}^2/m^2$	0.6459	0.2509	0.1290	0.0007	0.410	0.161
contact couplings alone ($g_0^{\text{eff}} = -g_{VS}^2/m^2$, $g_1^{\text{eff}} = +0.1\, g_{VS}^2/m^2$)	0.0006	0.0000	0.0006	0.0000	0.018	-0.018

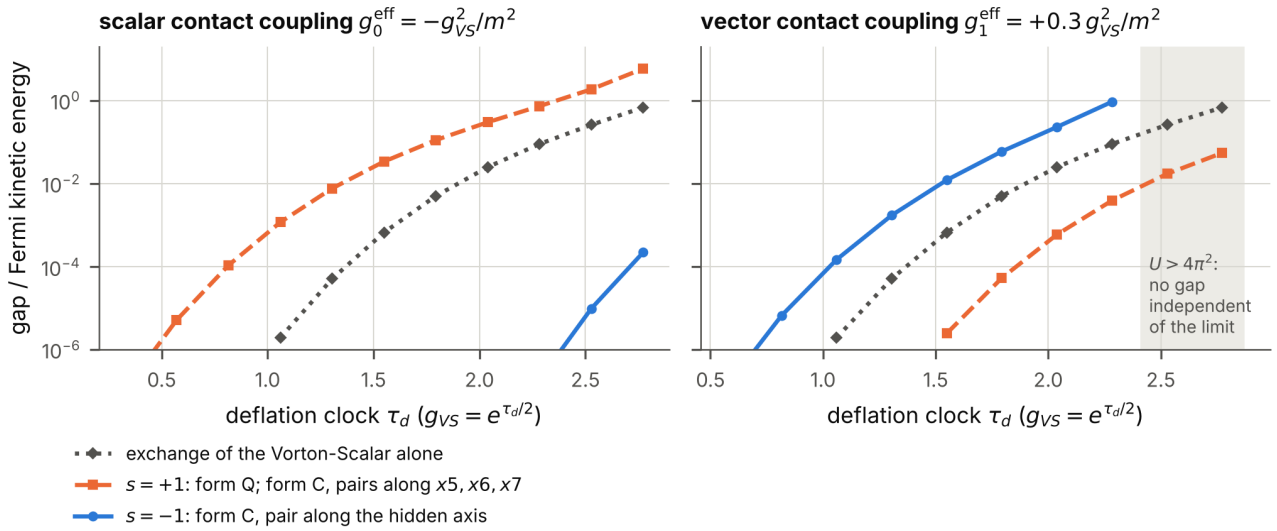


Figure 5: The gap at the Fermi surface, in units of the Fermi kinetic energy, along the deflation clock, for $m_s = m$ and $k_F = 0.5\, m$. Left: with the scalar contact coupling $g_0^{\text{eff}} = -g_{VS}^2/m^2$; right: with the vector contact coupling $g_1^{\text{eff}} = +0.3\, g_{VS}^2/m^2$. Dotted line: the exchange of the Vorton-Scalar alone. Dashed line (squares): the contact coupling with the sign $s = +1$ (form Q; form C in the pairs along $x5, x6, x7$). Solid line (circles): with the sign $s = -1$ (form C in the pair along the hidden axis). In the shaded band the kernel with $s = -1$ exceeds the ultraviolet criterion of Section 7.6, and no gap is given.

With $s = +1$ a scalar contact attraction raises the gap and a vector contact repulsion lowers it (Figure 5). With $s = -1$ (the hidden-axis pair of form C) the same couplings act in the opposite way: the scalar coupling lowers the gap (for $m_s = m$ it removes it at this clock reading), and the vector coupling raises it (checks `pc-P4`, `pc-P6`; the gap with $s = -1$ and the vector coupling agrees with an independent solution of the gap equation to 7.8×10^{-5}). Which of the two kinds of pairs has the larger gap, over the clock readings at which both gaps are defined:

interaction	readings at which the pairs along $x5, x6, x7$ have the larger gap	readings at which the hidden-axis pair has
with $g_0^{\text{eff}} = -g_{VS}^2/m^2$	53	0
with $g_0^{\text{eff}} = -g_{VS}^2/m^2$, $g_1^{\text{eff}} = +0.1\, g_{VS}^2/m^2$	53	0
with $g_1^{\text{eff}} = +0.3\, g_{VS}^2/m^2$	0	43
with $g_7^{\text{eff}} = -0.1\, g_{VS}^2/m^2$	51	0

In form C the self-interaction removes the degeneracy of the pair along the hidden axis with the pairs along $x5, x6, x7$, and so selects the direction of the order parameter. With an attractive scalar coupling the pairs along the extra time-like axes $x5, x6, x7$ have the larger gap, not the pair along the hidden axis. With a repulsive vector coupling alone the pair along the hidden axis has the larger gap. In form Q, as on 7+1, the four pairs have the same gap, and the direction is free.

7.6 The ultraviolet criterion of the gap equation

At momenta far above all masses every term of the pairing potential has the form (coupling) $\times$ (function of $\cos\theta$) $/q^2$, with $F_0, F_S, F_P \to \frac{1}{2}(1-\cos\theta)$, $F_8 \to -\frac{1}{2}(1-\cos\theta)$ and $F_1, F_7 \to 3-\cos\theta$. The kernel is then homogeneous, $\mathsf{W}(k,k') \to \mathsf{w}(k'/k)/k^2$, and above the gap Eq. (7.6) reads $\Delta(k) = \frac{1}{4\pi^2}\int_0^\infty dt\, t\, \mathsf{w}(t)\, \Delta(tk)$. A power $\Delta \propto k^{-\mathsf{s}}$ solves it if

$$J(\mathsf{s}) \equiv \int_0^\infty dt\; t^{1-\mathsf{s}}\, \mathsf{w}(t) = 4\pi^2. \tag{7.7}$$

Because W is symmetric in its arguments, $\mathsf{w}(1/t) = t^2\, \mathsf{w}(t)$ and $J(2-\mathsf{s}) = J(\mathsf{s})$: the two exponents are $\mathsf{s} = 1 \pm \sigma$, and J is smallest at $\mathsf{s} = 1$. Define the **ultraviolet strength**

$$U = J(1) = \int_0^\infty dt\; \mathsf{w}(t). \tag{7.8}$$

- If $U < 4\pi^2$ the exponents are real, the gap function falls as $k^{-1-\sigma}$, and the gap equation has a solution that does not depend on the upper limit K; it is approached as $K^{-2\sigma}$, slowly when U is close to $4\pi^2$.
- If $U > 4\pi^2$ the exponents are complex, no such solution exists, and with an upper limit the gap grows in proportion to K: the pair collapses to short distances. This is the fall to the center of relativistic particles in a potential $-\alpha/r$ with α above its critical value. The Fourier transform of $g/(q^2+\mu^2)$ is the Yukawa potential $g\, e^{-\mu r}/(4\pi r)$, which is a Coulomb potential at distances below $1/\mu$.

The integral of one term is elementary. With $\int_0^\infty dt/(1+t^2-2t\cos\theta) = (\pi-\theta)/\sin\theta$,

$$\int_0^\infty dt\; \frac{1}{2}\int_{-1}^{1} dx\; \frac{a+b\,x}{1+t^2-2tx} = \frac{\pi^2}{4}\, a + b, \tag{7.9}$$

that is $I_S = \pi^2/8 - \tfrac{1}{2} = 0.7337$ for $\tfrac{1}{2}(1-x)$ and $I_V = 3\pi^2/4 - 1 = 6.4022$ for $3-x$. Therefore

$$\boxed{U = g_{VS}^2\Big(\frac{\pi^2}{4}-1\Big) - s\, m_{\rm ct}^2\Big[(g_0^{\rm eff}-g_8^{\rm eff})\Big(\frac{\pi^2}{8}-\frac{1}{2}\Big) + (g_1^{\rm eff}+g_7^{\rm eff})\Big(\frac{3\pi^2}{4}-1\Big)\Big]} \tag{7.10}$$

when both modes of the Vorton-Scalar are exchanged (half the first term for the radial mode alone). The formula is verified against the numerical integral of the kernel (unit test of the program; check `Icks-F8` of the independent script). Per unit g_{VS}^2, with the contact couplings of the table of Section 7.5:

interaction	$s=+1$: U/g_{VS}^2	critical g_{VS}^2	τ_d	$s=-1$: U/g_{VS}^2	critical g_{VS}^2	τ_d
exchange of the Vorton-Scalar alone	1.467	26.90	3.29	1.467	26.90	3.29
with $g_0^{\rm eff} = -g_{VS}^2/m^2$	2.201	17.94	2.89	0.734	53.81	3.99
with $g_0^{\rm eff} = -g_{VS}^2/m^2$, $g_1^{\rm eff} = +0.1\, g_{VS}^2/m^2$	1.561	25.29	3.23	1.374	28.73	3.36
with $g_1^{\rm eff} = +0.3\, g_{VS}^2/m^2$	-0.453	—	—	3.388	11.65	2.46
with $g_7^{\rm eff} = -0.1\, g_{VS}^2/m^2$	2.108	18.73	2.93	0.827	47.73	3.87
contact couplings alone ($g_0^{\rm eff} = -g_{VS}^2/m^2$, $g_1^{\rm eff} = +0.1\, g_{VS}^2/m^2$)	0.093	422.32	6.05	-0.093	—	—

The exchange of the two modes alone reaches $U = 4\pi^2$ at $g_{VS}^2 = 4\pi^2/(\pi^2/4-1) = 26.90$, the clock reading 3.29, beyond the last reading 2.77 of the gap histories of this folder. The vector contact coupling $+0.3\, g_{VS}^2/m^2$ in the hidden-axis pair of form C reaches it at $g_{VS}^2 = 11.65$, the clock reading 2.46; from there on no gap is given (8 clock readings of the gap history, check `pc-P6`).

Numerical test (check pc-P7). The gap was computed with the upper limits $K = 60$, 2×10^3, 2×10^4 and 2×10^5 (in units of the larger of m and k_F) in 192 cases. Examples (gap in units of the Fermi kinetic energy; $m_s = m$, $k_F = 0.5\,m$):

interaction	τ_d	$U/4\pi^2$	$K = 60$	2×10^3	2×10^4	2×10^5
exchange of the Vorton-Scalar alone	2.04	0.285	0.024647	0.024674	0.024674	0.024674
exchange of the Vorton-Scalar alone	2.53	0.466	0.25708	0.25804	0.25805	0.25805
exchange of the Vorton-Scalar alone	2.77	0.595	0.66928	0.67618	0.67629	0.67629
with $g_0^{\rm eff} = -g_{VS}^2/m^2$, $s=+1$	2.04	0.428	0.29679	0.29751	0.29752	0.29752
with $g_0^{\rm eff} = -g_{VS}^2/m^2$, $s=+1$	2.53	0.698	1.8061	1.8486	1.8498	1.8500
with $g_0^{\rm eff} = -g_{VS}^2/m^2$, $s=+1$	2.77	0.892	5.0760	5.7318	5.8068	5.8262
with $g_1^{\rm eff} = +0.3\,g_{VS}^2/m^2$, $s=-1$	2.04	0.659	0.22460	0.22842	0.22849	0.22850
with $g_1^{\rm eff} = +0.3\,g_{VS}^2/m^2$, $s=-1$	2.53	1.075	4.8524	14.617	51.803	753.50
with $g_1^{\rm eff} = +0.3\,g_{VS}^2/m^2$, $s=-1$	2.77	1.373	34.748	1044.3	1.1991×10^4	1.3779×10^5

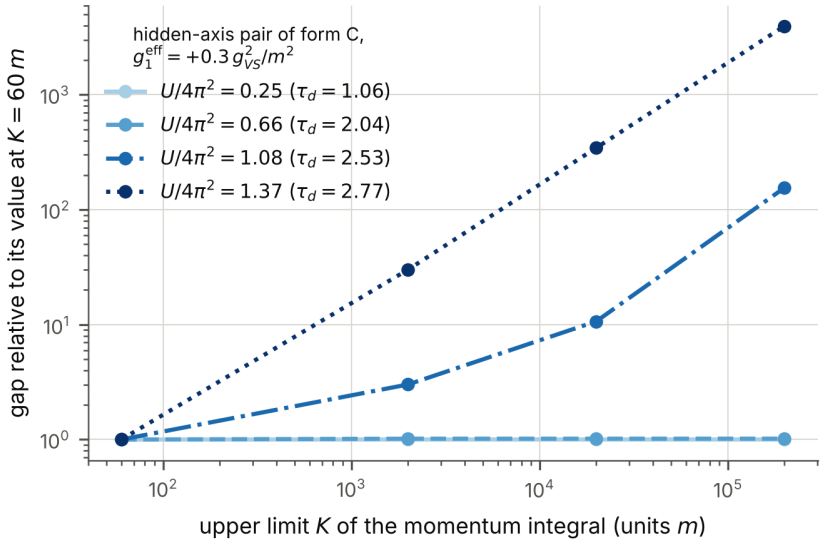


Figure 6: The gap as a function of the upper limit K of the momentum integral, relative to its value at $K = 60\,m$, for the hidden-axis pair of form C with the vector contact coupling $+0.3\,g_{VS}^2/m^2$ ($m_s = m$, $k_F = 0.5\,m$) at four clock readings. Below the criterion ($U/4\pi^2 < 1$, the two lower curves, which coincide on this scale) the gap does not depend on the limit; above it the gap grows with the limit without bound.

By classes of the ultraviolet strength, for the cases below the criterion:

ultraviolet strength	cases	change of the gap from $K = 2\times10^4$ to 2×10^5 (relative)	error of the limit $K = 60$ (per cent)
$U/4\pi^2 < 0.5$	118	2.5×10^{-6}	0.75
0.5 to 0.7	28	6.4×10^{-5}	2.37
0.7 to 0.9	8	3.3×10^{-3}	12.59

Above the criterion (8 cases) the gap grows by a factor between 11.49 and 14.56 from $K = 2 \times 10^4$ to 2×10^5; its value then depends on the limit and on the grid of momenta, and is not a property of the model. **All gaps of this folder are computed with $K = 2 \times 10^4$, and none is given above the criterion.** The earlier calculations of the project used $K = 60$; their gaps of the exchange of the Vorton-Scalar alone are too small by up to 1.08 per cent at the strongest coupling ($g_{VS} = 4$), and their gaps with contact couplings by the percentages of the last column.

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8. Symmetries

8.1 The frame group

The 28 matrices $\frac{1}{2}\gamma^A\gamma^B$ ($A < B$) generate the rotations and boosts of the 4+4 frame. The invariants I_r^{C} of form C are invariant under all 28 (check `44c-S4`). The positive kinetic and mass terms (2.5) are invariant only under the 13 generators that commute with W: the ten rotations and boosts of $x0, \dots, x4$ and the three rotations of $x5, x6, x7$. The other fifteen generators mix a space-like coordinate or the ordinary time with an extra time; they do not preserve the quantized sector (a field that does not depend on $x5, x6, x7$ is turned into one that does).

The invariants I_r^{Q} of form Q are invariant under the 28 matrices $\frac{1}{2}\gamma^A_{44q}\gamma^B_{44q}$, which generate the group of a spacetime with one time. Thirteen of them are the thirteen transformations just named; the other fifteen mix the two halves of the field and are not transformations of the coordinates of the 4+4 spacetime (check `44q-S4`).

Both models, as wholes, have the same thirteen symmetries of the coordinates. The difference is in the quartic term alone: that of form C is invariant under the full group of the 4+4 frame, as the hypothesis of the project requires of "the invariants of the spacetime"; that of form Q is invariant under a different group of the same size, which is not a symmetry group of the 4+4 spacetime.

8.2 The internal symmetry of fields in ordinary space

A field that depends on $x1, x2, x3, x4$ only has a kinetic term that contains $\alpha^1, \alpha^2, \alpha^3$ and β. Eight Hermitian matrices commute with these four: the six generators $\propto \gamma^a\gamma^b$ ($a < b$ in $\{0, 5, 6, 7\}$) of the rotations of the four extra axes, which form $SO(4)$; the unit matrix (the fermion number); and $P_E \propto \gamma^0\gamma^5\gamma^6\gamma^7$, which generates a second phase symmetry called $U(1)_E$. The model with the Vorton-Scalar and without contact self-interaction is invariant under all eight. The three rotations of $x5, x6, x7$ among themselves commute with W; the three generators $\gamma^0\gamma^t$ and P_E anticommute with W and connect the two halves.

Under which couplings is the self-interaction invariant under the unitary transformation $\psi \to e^{i\theta T}\psi$ generated by each of them? In exact arithmetic (checks `71s-S12`, `44q-S12`, `44c-S12`; confirmed at a finite angle by the independent script, check `Icks-F2`):

generators	number	$7+1$	$4+4$, form Q	$4+4$, form C
phase of the field (fermion number)	1	all couplings	all couplings	all couplings
rotations of $x5, x6, x7$ among themselves	3	all couplings	all couplings	all couplings
rotations that mix $x0$ with $x5$, $x6$ or $x7$	3	all couplings	all couplings	no coupling ($g_0^{\mathrm{eff}} = g_1^{\mathrm{eff}} = g_7^{\mathrm{eff}} = g_8^{\mathrm{eff}} = 0$)
P_E, the generator of $U(1)_E$	1	$g_1^{\mathrm{eff}} = g_7^{\mathrm{eff}} = 0$	$g_1^{\mathrm{eff}} = g_7^{\mathrm{eff}} = 0$	no coupling ($g_0^{\mathrm{eff}} = g_1^{\mathrm{eff}} = g_7^{\mathrm{eff}} = g_8^{\mathrm{eff}} = 0$)

In form C every self-interaction that does not vanish breaks the three transformations that mix the hidden axis with $x5, x6, x7$, and breaks $U(1)_E$. The internal symmetry of fields in ordinary space is reduced from $SO(4)\times U(1)\times U(1)_E$ to $SO(3)_T\times U(1)$: the rotations of the three extra time-like coordinates among themselves, and the fermion number.

The four conditions of the table for form C, $5g_0^{\text{eff}}-16g_7^{\text{eff}}+g_8^{\text{eff}}=0$, $5g_1^{\text{eff}}-4g_7^{\text{eff}}-g_8^{\text{eff}}=0$, $4g_7^{\text{eff}}+g_8^{\text{eff}}=0$ and $g_7^{\text{eff}}-g_8^{\text{eff}}=0$, have only the solution in which all four effective couplings vanish.

Why. In the frame group of the 4+4 spacetime $\gamma^0\gamma^t$ generates a **boost** (it mixes a space-like with a time-like coordinate), which is not a unitary transformation of the field. The *unitary* transformation $e^{i\theta\gamma^0\gamma^t}$ is a different transformation: it is a symmetry of the kinetic term of fields in ordinary space by accident, and it is not an element of the frame group. The quartic term of form C is invariant under the boost and has no reason to be invariant under the unitary transformation. In form Q, as on the 7+1 spacetime, where all four extra axes are space-like, the same matrices generate rotations, which are unitary, and the quartic term is invariant under them.

What this means for the pairs. It is the symmetry statement behind Section 7: in form Q the pair along $x0$ can be rotated into the pairs along $x5, x6, x7$, so that they are degenerate; in form C it cannot, and they are not.

8.3 The chiral condition

Under $\psi\to e^{i\epsilon\chi}\psi$ the self-interaction is invariant if and only if

$$g_0^{\text{eff}}+g_8^{\text{eff}}=0 \quad \text{(form Q, and 7+1)}, \qquad g_1^{\text{eff}}-g_7^{\text{eff}}=0 \quad \text{(form C)} \tag{8.1}$$

(checks `44q-S10`, `44c-S10`). The roles of the even and the odd ranks are exchanged, because the bilinears of the two forms differ by the matrix W, which anticommutes with χ. A scalar coupling alone ($g_0\neq 0$) is chirally invariant in form C and not in form Q; a vector coupling alone is chirally invariant in form Q and not in form C. The mass term breaks the chiral symmetry in both forms.

8.4 Consequence for the hypothesis that the pair condensate is the Higgs field

One of the author's hypotheses identifies the pair condensate along the hidden axis with the Higgs field: the six generators of the $SO(4)$ of Section 8.2 would play the role of the electroweak group, and the condensate along $x0$ would break it to the $SO(3)$ of the rotations of $x5, x6, x7$, as the Higgs vacuum breaks the electroweak group.

In form Q this pattern is present in the sector of fields in ordinary space for any couplings of the self-interaction, as a hidden symmetry (it is not a symmetry of the 4+4 spacetime). In form C it is not: the self-interaction itself breaks $SO(4)$ to $SO(3)_T$ *explicitly*. The three generators that the condensate would have to break spontaneously are not symmetries of the model; no massless modes of the order parameter correspond to them; and by Section 7 the condensate along $x0$ is not degenerate with those along $x5, x6, x7$. The pattern survives in form C only in the model without contact self-interaction (the exchange of the Vorton-Scalar keeps the whole internal symmetry).

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9. The Vorton-Scalar

The Yukawa coupling of the complex scalar field ϕ of the project to the fermions contains $\bar\psi\psi$ and $\bar\psi\, i\chi\,\psi$ with $\bar\psi=\psi^\dagger\beta$. Under the chiral rotation $\psi\to e^{i\epsilon\chi}\psi$ these two bilinears rotate into each other, and the coupling is invariant when the phase of ϕ rotates with them; the phase mode of the scalar is the Goldstone mode of this symmetry. The covariant bilinears cannot take this role: C^-

commutes with χ, so that $\tilde{\psi}\psi$ and $\tilde{\psi}\,\chi\,\psi$ are each invariant under the chiral rotation and do not couple to the phase of a complex field. The Yukawa coupling is therefore written with $\bar{\psi}$ in both forms, and an expectation value of the scalar shifts the masses of both halves by the same amount.

With a bare Dirac mass m_D, a Yukawa mass m_Y ($m = m_D + m_Y$ in vacuum, $f_D = m_D/m$) and the self-interaction, the energy density of uniform matter is

$$\varepsilon = \frac{(1-f_D)^2\,(x^2-1)^2}{8\,C} + \varepsilon_{\text{halves}}\big[m \to f_D + (1-f_D)\,x\big], \tag{9.1}$$

where $x = \phi/v_{VS}$ is the expectation value of the scalar in units of its vacuum value, $C = g_{VS}^2 m^2/m_s^2$ its coupling, and $\varepsilon_{\text{halves}}$ the energy density (5.9) with the mass m replaced as shown (units $m = 1$, $D = 3$). The smallest C at which matter is self-bound, with the contact couplings $g_0 = -\kappa_0\,C$ and $g_1 = +\kappa_1\,C$, is ($\Delta\tau_d$: the shift of the clock reading against the model without bare mass and self-interaction, where $C = 3\pi^2/8$; x: the value of the scalar field in the bound state just above the threshold):

f_D	κ_0	κ_1	form C: C ($\Delta\tau_d$)	x	form Q: C ($\Delta\tau_d$)	x
0	0	0	3.7011 (0)	0.000	3.7011 (0)	0.000
0	1	0	3.9399 (+0.063)	0.000	3.7156 (+0.004)	0.614
0	1	0.1	3.9399 (+0.063)	0.000	4.1527 (+0.115)	0.652
0.25	0	0	2.5112 (−0.388)	0.000	2.5112 (−0.388)	0.000
0.25	1	0	2.6336 (−0.340)	0.000	2.4758 (−0.402)	0.000
0.25	1	0.1	2.6307 (−0.341)	0.000	2.6992 (−0.316)	0.000
0.5	0	0	2.0643 (−0.584)	0.000	2.0643 (−0.584)	0.000
0.5	1	0	2.1617 (−0.538)	0.000	1.8252 (−0.707)	0.000
0.5	1	0.1	2.1559 (−0.540)	0.000	1.9335 (−0.649)	0.000

In all 18 cases the bound state at the threshold has equal halves, in both forms: the attraction that binds it is that of the scalar field, which acts on both halves alike, and the contact couplings of these strengths are too weak to polarize it. The threshold is that of matter of one kind with the couplings T_S, T_V of the form, confirmed with the functional of two halves minimized over the density, the field and the polarization (bound 0.1 per cent above, with $\varepsilon/n - m$ at least $1.6 \times 10^{-4}\ m$ above zero 0.1 per cent below; check `fm-F8`). The forms differ because the couplings of the total densities differ (table of Section 5.2). In form Q the scalar contact coupling adds its attraction G_S^{HF} to that of the scalar field, and a vector coupling adds the repulsion G_V^{HF}. In form C the couplings of the total densities are G_S^{c} and G_V^{c}, both small, so that the contact couplings change the threshold less. Which form binds earlier depends on the case. Without bare mass the bound state of form C at the threshold has $x = 0$: the fermions are massless in it, the scalar densities vanish, and the threshold does not depend on the vector coupling of set B.

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10. Gravity

The equations. Let the three rates be equal, g55 = g66 = g77, and let the four space-like coordinates have a scale factor of their own:

$$ds^2 = -(dx4)^2 + \mathsf{a}(x4)^2 \sum_{j=0}^{3} (dxj)^2 - \mathsf{b}(x4)^2 \sum_{t=5}^{7} (dxt)^2, \qquad H_{\mathsf{a}} = \frac{\mathsf{a}'}{\mathsf{a}}, \quad H_{\mathsf{b}} = \frac{\mathsf{b}'}{\mathsf{b}} \tag{10.1}$$

(a prime is a derivative with respect to $x4$). The deflating spacetime of the hypothesis is $\mathsf{a} = 1$, $H_{\mathsf{b}} = -\mathrm{g55}$; it is called the *undisturbed deflation*. For such a metric the curvature is determined by one number for each pair of directions of the frame,

$$K_{4i} = H_i' + H_i^2, \qquad K_{ij} = H_i\, H_j \quad (i \neq j), \tag{10.2}$$

where i, j run over the seven directions other than the time and H_i is the rate of the direction i. The field equations of second order for the metric in eight dimensions are the Einstein–Lovelock equations,

$$\sum_{k=1}^{3} \alpha_k\, E^{(k)\mu}{}_{\nu} - \bar{\Lambda}\, \delta^{\mu}_{\nu} = \kappa_8\, T^{\mu}{}_{\nu}, \qquad \alpha_1 = 1, \tag{10.3}$$

$$E^{(k)m}{}_{m} = (-1)^{k+1}\, k!\, 2^{k-1} \sum_{\text{sets of } 2k \text{ directions} \neq m} \; \sum_{\text{divisions into pairs}} \; \prod_{\text{pairs}} K_{\text{pair}} \tag{10.4}$$

(no summation over m; $E^{(1)}$ is minus the Einstein tensor), with the couplings α_2, α_3 of the two higher terms, the cosmological term $\bar{\Lambda}$ and the gravitational coupling κ_8. The energy-momentum tensor of uniform matter has $T^4{}_4 = \varepsilon$ and $T^m{}_m = -p_m$, with the pressure p of Eq. (5.10) along $x0, \ldots, x3$ and the interaction pressure Π along $x5, x6, x7$. Without matter the undisturbed deflation solves Eq. (10.3) if and only if

$$\bar{\Lambda} = 3\, \mathrm{g55}^2, \qquad \alpha_2 = \frac{1}{4\, \mathrm{g55}^2} \tag{10.5}$$

(α_3 arbitrary): the *tuned* Einstein–Lovelock gravity. *Einstein gravity* is $\alpha_2 = \alpha_3 = 0$, $\bar{\Lambda} = 0$, with the matter as the only source. The **loading** of an evolution is the number $\kappa_8\, \varepsilon(0)/(3\, \mathrm{g55}^2)$, the energy density of the matter at the start in the unit that the deflation sets.

The matter of the two forms as the source (fields of the quantized sector, $D = 4$, set B, three times the binding threshold of each form):

form	threshold $g\, m^3$	state at 3 times the threshold	k_F/m of the filled states	M/m	$\varepsilon/n - m$ (units m)
C	332.2917	one half filled	0.52859	0.37920	−0.09072
Q	121.0858	equal halves	0.58171	0.35844	−0.10601

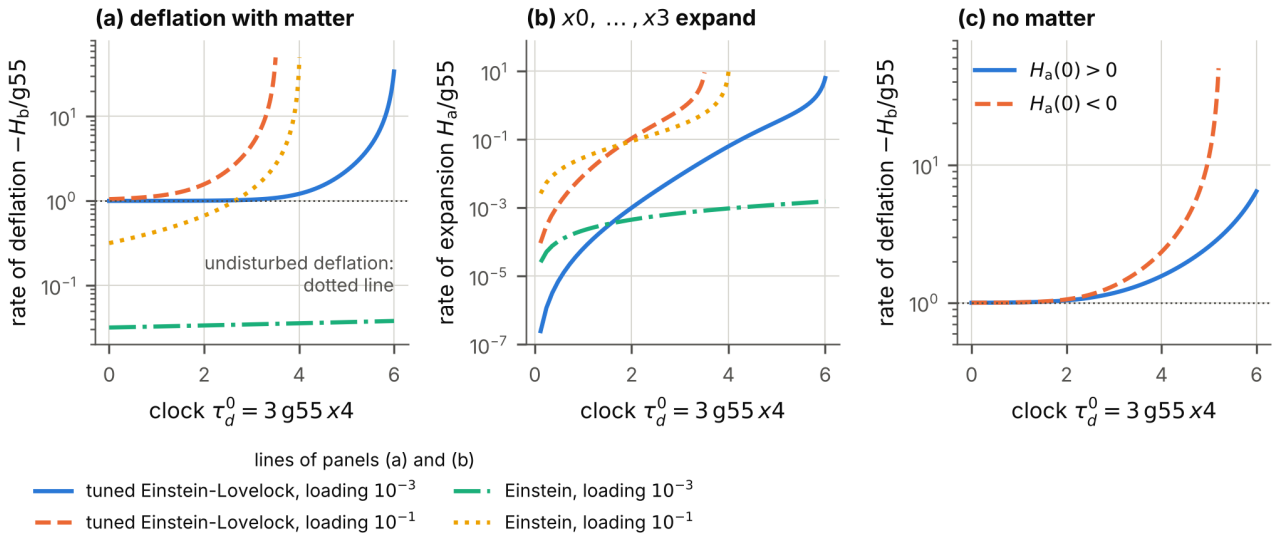


Figure 7: The deflating 4+4 spacetime with the matter of form C (one half filled) as a source of gravity. (a) The rate of deflation $-H_{\rm b}$ of $x5, x6, x7$ in units of its undisturbed value g55, along the clock $3\,{\rm g55}\,x4$ of the undisturbed deflation; (b) the rate $H_{\rm a}$ at which the coordinates $x0, \dots, x3$ expand; four evolutions: the tuned Einstein–Lovelock gravity and Einstein gravity, each with the loadings 10^{-3} and 10^{-1}. (c) The same spacetime without matter in the tuned gravity, slightly disturbed. The deflation does not return to its undisturbed rate in any of the evolutions.

In an evolution without collisions every particle keeps its state, so that a half that is empty stays empty. The matter of form C is then matter of one kind of particle with four internal states and the couplings a_S, a_V throughout the evolution; the matter of form Q is matter of one kind with eight states and the couplings $G_S^{\rm HF}$, $G_V^{\rm HF}$. Both were computed again with the routines for matter of one kind (other integrals), with agreement to 1.3×10^{-15} (check `gr-G6`). In the evolutions (Figure 7 for form C) the two kinds of matter behave alike in kind: the conclusions on the stability of the deflation do not depend on the form. With the tuned Einstein–Lovelock gravity and the loading 10^{-3} the rate of the deflating coordinates at the end of the evolution is 34.1 times its initial value for the matter of form C and 15.6 times for the matter of form Q.

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11. Discussion of the two options

11.1 What each option is

Both options keep what the quantum theory needs: the positive kinetic term, a real (Hermitian) interaction, and the conservation of the number of particles. Both use the same nine couplings and the same four effective couplings. They assign to these couplings different meanings. Form C takes the hypothesis literally: the quartic term is built from the invariants of the 4+4 spacetime. Form Q takes the *particle content* literally: the eight internal states of a particle are treated alike, as on the 7+1 spacetime.

11.2 Arguments for form C

1. **It is the quartic term of the hypothesis.** The invariants $I_r^{\rm C}$ are the ones that are invariant under all rotations and boosts of the 4+4 frame. They can be written on the full deflating spacetime, for fields that depend on all eight coordinates, with the spin connection; the quantized sector is a restriction of that covariant theory. Form Q is defined in the quantized sector only:

its invariance group is not a symmetry group of the 4+4 spacetime, and it has no covariant extension to fields that depend on the extra times.

2. **Its dynamics respects the distinction that the metric makes.** On the 4+4 spacetime $x0$ is space-like and $x5, x6, x7$ are time-like. Form C treats them differently (Sections 7 and 8); form Q has an accidental symmetry that rotates them into one another.
3. **Each structure appears with its own adjoint.** The mass that the interaction of form C generates in vacuum is the covariant mass term (Section 6.2).

11.3 Arguments against form C, or for form Q

1. **The two adjoints are mixed.** Form C combines the quartic term of the covariant Lagrangian with the kinetic and mass terms of the positive Lagrangian. The Lagrangian as a whole is invariant only under the thirteen generators that both parts share, the same thirteen as for form Q. The covariance of the quartic term is not a symmetry of the model.
2. **The bare mass term and the dynamical mass have different structures** in form C. With a bare mass the vacuum has a sharp transition at which the symmetry between the halves breaks spontaneously (Section 6.3). In form Q the two have the same structure.
3. **The results of the 7+1 spacetime do not carry over.** In form Q every result for fields that do not depend on $x5, x6, x7$ is the same on the two spacetimes; in form C the 4+4 model is a different model, with polarized matter, later binding, and pairs that are not degenerate.
4. **The pattern of the Higgs hypothesis is lost** as soon as there is a contact self-interaction (Section 8.4), and with an attractive scalar coupling the condensate does not form along the hidden axis (Section 7.5).
5. **A scalar coupling of the sign that binds matter on 7+1 binds less.** Matter needs two to three times the strength (Section 5.5), which moves it further outside the range in which the no-sea approximation is controlled (Section 6.4).

11.4 What does not distinguish them

- The reduction to four effective couplings, the Fierz matrix and all algebra of Section 4.3.
- The interaction between particles of the same half, hence everything about matter that occupies one half only.
- The pairs along $x5, x6, x7$.
- The exchange of the Vorton-Scalar, hence every result of the project that does not involve the contact self-interaction.
- The equivalence of the Kohn–Sham and the CJT method.
- The conclusions on gravity (Section 10).
- The critical strength at which the vacuum of a massless field becomes unstable, and the magnitude of the mass that it acquires.

11.5 What could decide

Three kinds of consideration could decide between the options; none of them is settled by the calculations of this folder.

1. *A principle.* If the model is required to be the restriction of a covariant theory on the full 4+4 spacetime, form C follows. If it is required to be the theory of eight equivalent internal states, form Q follows.
2. *A consistent quantization of the covariant Lagrangian itself.* If a quantization of Eq. (2.1) with its indefinite sign existed (with an indefinite metric on the space of states and a rule for the physical states), it would dictate the form of the interaction, and it would give neither form C nor form Q in general (Section 3.4). No such quantization is constructed in this project.
3. *An observable consequence.* The two forms differ in the internal states of self-bound matter (four or eight), in the direction of the condensate, and in the symmetry of the order parameter. A hypothesis of the project that ties one of these to an observation would select a form.

---

12. What has been established, and what has not

Established (exact algebra, verified by the program in integer arithmetic and independently by a second program):

- the two forms are related by the insertion of W in every bilinear (4.1), (4.2);
- the five linear relations among the nine invariants and the four effective couplings are the same in both forms;
- in matter without coherence between the halves the Hartree–Fock energy is the sum of four squares (5.3), with the same four combinations of couplings in both forms and the roles of the totals and the differences exchanged;
- the pair couplings of the 28 pairs at rest, and the sign rule (7.4) of the contact amplitudes at all momenta;
- the conditions of invariance under the internal symmetry and under the chiral rotation, in both forms; in particular that every self-interaction of form C breaks $SO(4) \times U(1)_E$ to $SO(3)_T$;
- the ultraviolet criterion $U < 4\pi^2$ of the gap equation and the formula (7.10).

Established numerically (each by two methods or two programs):

- matter of form C is polarized, and self-bound matter has one filled half; matter of form Q is not polarized; the binding thresholds and the saturated states along the clock for the sets A and B;
- the vacuum of a massless field acquires masses of opposite sign in the two halves in form C and of equal sign in form Q, at the same critical strength; with a bare mass form C has a sharp transition;
- the gaps of the two kinds of pairs with contact couplings, and their dependence on the upper limit of the momentum integral.

Not established:

- *States with coherence between the halves.* The density matrix (5.1) is diagonal in W. Because the interaction connects the halves, states in which $\langle \psi_+^\dagger \cdot \psi_- \rangle \neq 0$ are possible in principle, in both forms; whether one of them has a lower energy than the states found here has not been examined. The statements of Section 5 are statements about the states of the form (5.1).
- *Other couplings.* The numbers are for the sets A and B (scalar attraction, with and without a small vector repulsion). The exact formulas hold for all couplings; the numerical statements have not been extended to couplings with g_7 or g_8.
- *Beyond the mean field.* All results on matter and on the vacuum are in the Hartree–Fock approximation (equivalently, the CJT effective action in the double-bubble truncation), and those on matter in the no-sea approximation, which at the binding thresholds is outside its controlled range (Section 6.4).
- *Solutions with a negative mass.* Where the gap equations of matter have no solution with non-negative masses (Section 5.4), the continuation with a mass that has changed its sign has not been followed.
- *Which form is right.* Section 11.5.

---

13. Where the two forms are recorded, and how to repeat every number

13.1 In the program

The program of this folder is the Rust crate `code/deflation-cjt-ks-c-rs` (binary `deflation_cjt_ks_c`). Nothing of form Q was removed from it; form C stands beside it and is labeled as the definition.

object	form C	form Q	source file
the 256 matrices, exact algebra	model `S44C`, tag `44c`	model `S44Q`, tag `44q`	`src/selfint.rs`
mean-field couplings of two halves	`Form::C`	`Form::Q`	`src/halves.rs`
matter of two halves, vacuum, functional with the scalar	`Halves`, `CjtHalves`, `UnifiedHalves` with the form as a parameter	the same	`src/halves.rs`
sign of the contact couplings in a pair	`ct_sign = -1` (pairs along $x0, \dots, x3$), `+1` (pairs along $x5, x6, x7$)	`ct_sign = +1`	`src/pair_contact.rs`

command	what it computes for the two forms	check identifiers
`selfcoupling`	exact algebra of both forms; relation (4.1); Hartree–Fock couplings; pairs at rest; symmetry conditions	`44c-S1` to `44c-S12`, `44q-S1` to `44q-S12`, `44-S13`
`forms`	everything of Sections 5, 6 and 9	`fm-F1` to `fm-F9`
`pairing`	amplitudes and gaps of the two kinds of pairs; the ultraviolet criterion	`pc-P1` to `pc-P7`
`gravity`	the matter of the two forms as a source	`gr-G6`

The independent verification is the Python script `code/scripts/verify_cjt_ks_independent.py`, part F (checks `Icks-F1` to `Icks-F8`), which obtains the same results by other methods: Wick's theorem with numerical density matrices, the symmetry at a finite angle, projectors from numerical eigenvectors, a fixed-point solver of the gap equations, adaptive quadrature.

13.2 How to repeat

From the folder `Revision_Deflation_CJT_and_Kohn-Sham_definition_C`:

```
# 1. compile the program and run its unit tests
cd code/deflation-cjt-ks-c-rs
cargo build --release
cargo test --release
cd ../..
# 2. the four commands that concern the two forms (they write the files in data/)
code/deflation-cjt-ks-c-rs/target/release/deflation_cjt_ks_c selfcoupling
code/deflation-cjt-ks-c-rs/target/release/deflation_cjt_ks_c forms
code/deflation-cjt-ks-c-rs/target/release/deflation_cjt_ks_c pairing
code/deflation-cjt-ks-c-rs/target/release/deflation_cjt_ks_c gravity
# 3. the independent verification (all parts; part F concerns the two forms)
python3 code/scripts/verify_cjt_ks_independent.py
# 4. the tables of this document, and the document itself
python3 code/scripts/write_doc_tables.py
python3 code/scripts/assemble_docs.py docs_src/P3_two_forms_of_the_self_interaction.src.md
bash code/scripts/build_docs.sh provenance/P3_two_forms_of_the_self_interaction.md
```

Each command prints its checks and ends with the number of checks that passed.

13.3 The data files

file in data/	content	used in
selfcoupling_hartree_fock.csv	the four mean-field couplings in the nine couplings, for both forms	5.1
selfcoupling_pairs.csv	the pair couplings of all pair operators, with the behavior under W	7.2
selfcoupling_internal_symmetry.csv	the conditions of invariance under the internal symmetry	8.2
forms_couplings.csv	$T_V, T_S, D_V, D_S, a_V, a_S, b_V, b_S$ of the sets A and B	5.2
forms_energy_vs_polarization.csv	energy against polarization	5.4
forms_thresholds.csv	binding thresholds	5.5
forms_saturation.csv	saturated matter along the clock	5.6
forms_fixed_density.csv, forms_gas_events.csv	matter of fixed density	5.7
forms_cjt_vs_ks.csv	the two methods compared	5.8
forms_vacuum_critical.csv, forms_vacuum.csv	the vacuum	6
forms_unified_thresholds.csv	thresholds with the Vorton-Scalar	9
forms_reversed_sign.csv	a scalar coupling of the opposite sign (Appendix A)	A
pair_amplitudes_forms.csv	pairing amplitudes of both forms in both kinds of pairs	7.3
pair_gap_history.csv	the gaps along the clock for both signs, with $U/4\pi^2$	7.5
pair_gap_upper_limit.csv	the gaps for four upper limits of the momentum integral	7.6
gravity_deflating_summary.csv	the evolutions with the matter of the two forms	10
forms_checks.txt, selfcoupling_checks.txt, pairing_checks.txt, gravity_checks.txt, cjt_ks_checks_independent.txt	the reports of the checks	all

The figures of this document are drawn from these files by code/scripts/make_figures.py (functions fig_forms_polarization, fig_forms_saturation, fig_forms_gas, fig_forms_vacuum, fig_forms_gap, fig_forms_uv, fig_gravity_deflating_formC, fig_gravity_deflating_formQ); the script computes nothing except ratios for display. To draw them again:

```
python3 code/scripts/make_figures.py forms_polarization forms_saturation forms_gas \
        forms_vacuum forms_gap forms_uv gravity_deflating_formC gravity_deflating_formQ
```

13.4 Changing the definition

The choice between the forms is a label, not a change of the calculations: the program computes both. The folder Revision_Deflation_CJT_and_Kohn-Sham of the repository contains the same project with form Q as the definition; this folder contains it with form C as the definition. The numbers of form Q in this folder and of form C in that folder are computed by the same routines.

Appendix A. A scalar coupling of the opposite sign

For completeness the command forms also examines a scalar coupling of the sign that repels on 7+1, $g_0 > 0$ alone (check fm-F9). Its statement, verbatim:

> a scalar coupling of the opposite sign, g_0 > 0 alone. In the covariant form particles of different halves attract each other; the coupling of the square of the total density is T_V = G_Vc = -g_0/16 < 0, and the energy per particle of uniform matter falls without bound as the density grows: the contact model then has no saturated state without a cutoff. In the quantized-sector form the same coupling repels (T_S = 15 g_0/16 > 0) and the energy per particle rises with the density. D = 4, g_0 = +100 m^-3, form C: E/N - m = +0.089,

-1.518, -35.819, -641.710 m at k_F,total = 1, 2, 4, 8 m; D = 4, g_0 = +100 m^-3, form Q: E/N - m = +1.095, +15.250, +243.184, +3890.737 m at k_F,total = 1, 2, 4, 8 m; D = 3, g_0 = +15 m^-2, form C: E/N - m = +0.158, -0.054, -5.220, -58.160 m at k_F,total = 1, 2, 4, 8 m; D = 3, g_0 = +15 m^-2, form Q: E/N - m = +0.929, +6.205, +48.701, +389.107 m at k_F,total = 1, 2, 4, 8 m

---

Appendix B. Complete report of the checks of the command `forms`

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham_definition_C: checks of the calculations
 ↪  for the two forms of the self-interaction on the 4+4 spacetime, C = covariant form (the
 ↪  definition) and Q = quantized-sector form (the alternative): uniform matter, effective
 ↪  potential, vacuum, Vorton-Scalar (fm-..)
generated by: deflation_cjt_ks_c (Rust crate spacetimes-deflation-cjt-ks-c, version 0.1.0;
 ↪  PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham_definition_C); command:
 ↪  deflation_cjt_ks_c forms
9 of 9 checks passed

[PASS] fm-F1  (numerical) the mean-field couplings and the solver of two halves. (i) The four
 ↪  couplings G_S^HF = (15 g_0eff - 8 g_1eff - 8 g_7eff - g_8eff)/16, G_V^HF = (-g_0eff + 22
 ↪  g_1eff - 6 g_7eff + g_8eff)/16, G_Sc = (-g_0eff - 2 g_1eff + 2 g_7eff + g_8eff)/16, G_Vc =
 ↪  -(g_0eff + g_8eff)/16 computed from these closed forms equal those of the exact coefficient
 ↪  tables (checks 44c-S7, 44q-S7) for the sets of couplings used here: set A (scalar attraction
 ↪  alone: g_0 = -g, all other couplings zero): G_S^HF = -0.9375 g, G_V^HF = 0.0625 g, G_Sc =
 ↪  0.0625 g, G_Vc = 0.0625 g; set B (the ratio of the author's example: g_0 = -g, g_1 = +0.1 g,
 ↪  all other couplings zero): G_S^HF = -0.9875 g, G_V^HF = 0.2 g, G_Sc = 0.05 g, G_Vc = 0.0625 g.
 ↪  Covariant form C: T_V = G_Vc, T_S = G_Sc (total densities), D_V = G_V^HF, D_S = G_S^HF
 ↪  (differences); quantized-sector form Q: the two pairs exchanged. (ii) The solver of the
 ↪  coupled gap equations (closed forms of the momentum integrals; all solutions) against the
 ↪  Kohn-Sham routines for matter of one kind of particle (Gauss-Legendre integrals), for 48 sets
 ↪  of form, dimension, couplings and Fermi momentum: with equal halves the symmetric solution is
 ↪  matter of eight states with the couplings T_S, T_V (mass, energy density, pressure agree to
 ↪  2.0e-15); with one half filled it is matter of four states with the couplings a_S = T_S + D_S,
 ↪  a_V = T_V + D_V (2.2e-15). (iii) With unequal halves (k_- = 0.8 k_+) the chemical potentials
 ↪  mu_+- = sqrt(k_+-^2 + M_+-^2) + V_+- are the derivatives of the energy density with respect to
 ↪  the densities of the halves (central differences: 2.7e-10), and the pressure from the momentum
 ↪  integrals equals mu_+ n_+ + mu_- n_- - eps (6.7e-16)
[PASS] fm-F2  (numerical) the binding thresholds of the two forms for the same couplings of the
 ↪  Lagrangian (fields that do not depend on x5, x6, x7; no-sea approximation; state of lowest
 ↪  energy with respect to the polarization): the smallest strength g = |g_0| at which matter is
 ↪  self-bound, verified in each case (bound 0.1 per cent above, not bound 0.1 per cent below): D
 ↪  = 4, set A, form C: g m^3 = 224.7386 (one half; k_F,total = 0.6140, M_+ = 0.5239, M_- =
 ↪  1.5441); D = 4, set A, form Q: g m^3 = 82.1116 (equal halves; k_F,total = 0.8054, M_+ =
 ↪  0.4946, M_- = 0.4946); D = 4, set B, form C: g m^3 = 332.2917 (one half; k_F,total = 0.5211,
 ↪  M_+ = 0.5625, M_- = 1.4842); D = 4, set B, form Q: g m^3 = 121.0858 (equal halves; k_F,total =
 ↪  0.6792, M_+ = 0.5424, M_- = 0.5424); D = 3, set A, form C: g m^2 = 24.5717 (one half;
 ↪  k_F,total = 0.5374, M_+ = 0.6454, M_- = 1.4053); D = 3, set A, form Q: g m^2 = 9.8090 (equal
 ↪  halves; k_F,total = 0.7356, M_+ = 0.6276, M_- = 0.6276); D = 3, set B, form C: g m^2 = 31.1674
 ↪  (one half; k_F,total = 0.4635, M_+ = 0.6712, M_- = 1.3638); D = 3, set B, form Q: g m^2 =
 ↪  12.3819 (equal halves; k_F,total = 0.6351, M_+ = 0.6575, M_- = 0.6575). Where the bound state
 ↪  has equal halves or one half filled, the threshold equals the one computed for matter of one
 ↪  kind of particle by the routines of module kohn_sham (largest relative difference 5.7e-14).
 ↪  Ratio of the threshold of the covariant form to that of the quantized-sector form: D = 4, set
 ↪  A: 2.737; D = 4, set B: 2.744; D = 3, set A: 2.505; D = 3, set B: 2.517
```

[PASS] fm-F3 (numerical) the polarization of the two halves. For the same couplings of the Lagrangian (sets A and B, attractive scalar coupling g_0 < 0), 1.5, 3 and 6 times the strength at which the quantized-sector form first binds matter, and total densities of the Fermi momenta 0.4, 0.6, 0.8, 1.0 (48 cases): in the covariant form a particle is attracted by the particles of its own half and repelled by those of the other, and the state of lowest energy is polarized in 48 of the 48 cases; in the quantized-sector form it is polarized in 0 cases. Polarization (n_+ - n_-)/n of the state of lowest energy in the covariant form (four densities in the order given): D = 4, set A, 1.5 times: 1.00, 1.00, 1.00, 1.00; D = 4, set A, 3 times: 1.00, 1.00, 1.00, 1.00; D = 4, set A, 6 times: 1.00, 1.00, 1.00, 0.78; D = 4, set B, 1.5 times: 1.00, 1.00, 1.00, 0.35; D = 4, set B, 3 times: 1.00, 1.00, 0.65, 0.22; D = 4, set B, 6 times: 1.00, 1.00, 0.39, 0.19; D = 3, set A, 1.5 times: 1.00, 1.00, 1.00, 1.00; D = 3, set A, 3 times: 1.00, 1.00, 1.00, 1.00; D = 3, set A, 6 times: 1.00, 1.00, 1.00, 0.89; D = 3, set B, 1.5 times: 1.00, 1.00, 1.00, 0.45; D = 3, set B, 3 times: 1.00, 1.00, 0.81, 0.31; D = 3, set B, 6 times: 1.00, 1.00, 0.50, 0.22. Example: D = 4, set B, three times the threshold of the quantized-sector form, total density of the Fermi momentum 0.8 m: energy per particle minus m = +0.3287 m in the covariant form (polarization 0.65) and +0.1323 m in the quantized-sector form (polarization 0.00). Existence of the states: the energy was computed at 41 polarizations in each case and form (3936 states). In the covariant form every one of these states exists. In the quantized-sector form 108 states with unequal halves do not exist (D = 4, set A, form Q, 6 times, k_F,total = 1 m: 39 of 41; D = 4, set B, form Q, 3 times, k_F,total = 1 m: 2 of 41; D = 4, set B, form Q, 6 times, k_F,total = 0.8 m: 4 of 41; D = 4, set B, form Q, 6 times, k_F,total = 1 m: 31 of 41; D = 3, set A, form Q, 6 times, k_F,total = 1 m: 15 of 41; D = 3, set B, form Q, 6 times, k_F,total = 1 m: 17 of 41): the gap equations have no solution with non-negative masses, because the attraction by the more populated half (coupling b_S between the halves, stronger than the coupling a_S inside a half) would make the mass of the particles of the less populated half negative; this happens only at 3 or 6 times the threshold and k_F,total >= 0.8 m, and the states with equal halves and with one half filled exist in all cases

[PASS] fm-F4 (numerical) self-bound matter along the deflation clock in the two forms (clock counted from the moment at which matter of the respective form first binds; 101 clock readings from 0 to 5 in each of the 8 cases, the couplings growing to 148 times their value at the threshold): the energy per particle of the saturated state falls monotonically, its pressure vanishes (largest |p|/eps after the first two readings: 2.8e-8), and the kind of the state does not change: one half filled in the covariant form, equal halves in the quantized-sector form. D = 4, set A, form C (one half): E/N - m = -0.1092 m at tau_d = 1 and -0.2913 m at tau_d = 3 (mass of the particles of the filled half or halves 0.351 m and 0.183 m); D = 4, set A, form Q (equal halves): E/N - m = -0.1284 m at tau_d = 1 and -0.3438 m at tau_d = 3 (mass of the particles of the filled half or halves 0.319 m and 0.156 m); D = 4, set B, form C (one half): E/N - m = -0.0829 m at tau_d = 1 and -0.2209 m at tau_d = 3 (mass of the particles of the filled half or halves 0.392 m and 0.215 m); D = 4, set B, form Q (equal halves): E/N - m = -0.0968 m at tau_d = 1 and -0.2580 m at tau_d = 3 (mass of the particles of the filled half or halves 0.371 m and 0.199 m); D = 3, set A, form C (one half): E/N - m = -0.1226 m at tau_d = 1 and -0.3365 m at tau_d = 3 (mass of the particles of the filled half or halves 0.373 m and 0.157 m); D = 3, set A, form Q (equal halves): E/N - m = -0.1424 m at tau_d = 1 and -0.3945 m at tau_d = 3 (mass of the particles of the filled half or halves 0.347 m and 0.137 m); D = 3, set B, form C (one half): E/N - m = -0.0940 m at tau_d = 1 and -0.2567 m at tau_d = 3 (mass of the particles of the filled half or halves 0.407 m and 0.183 m); D = 3, set B, form Q (equal halves): E/N - m = -0.1093 m at tau_d = 1 and -0.2990 m at tau_d = 3 (mass of the particles of the filled half or halves 0.389 m and 0.170 m). Regularity: the largest eigenvalue of the response matrix of the two gap equations stays below 1 at all readings (largest value 0.881), so that each state is the regular continuation of the dilute gas. Beyond the table the saturated state of the quantized-sector form, which has equal halves, reaches the eigenvalue 1 (2 D_S ds/dM = 1: the two masses can split) at: D = 4, set A: tau_d = 6.416; D = 4, set B: not before tau_d = 9; D = 3, set A: tau_d = 5.296; D = 3, set B: tau_d = 7.323; the saturated state of the covariant form has one half filled and a negative eigenvalue throughout

[PASS] fm-F5 (numerical) matter of fixed density while the couplings grow with the deflation clock (clock counted from the binding threshold of the respective form; total densities of the Fermi momenta 0.3 m and 0.6 m; D = 4 and D = 3; sets A and B). In the covariant form the matter polarizes at a definite clock reading before the binding threshold: the two halves become unequal, and within at most 0.35 of the clock one half is empty; where the pressure along the coordinates of the motion becomes negative, it does so after the polarization, and in 3 of the 8 cases (the higher density) the pressure stays positive for -3 <= tau_d <= 5, because the repulsion a_V between the particles of the filled half grows with the couplings while the attraction saturates as the mass falls. In the quantized-sector form the halves stay equal and the pressure becomes negative in all 8 cases. D = 4, set A, form C, k_F,total = 0.3 m: halves unequal from tau_d = -0.52, one half empty from -0.17; pressure negative from tau_d = 0.31; D = 4, set A, form C, k_F,total = 0.6 m: halves unequal from tau_d = -1.77, one half empty from -1.42; pressure negative from tau_d = -0.43; D = 4, set A, form Q, k_F,total = 0.3 m: halves equal throughout; pressure negative from tau_d = 0.78; D = 4, set A, form Q, k_F,total = 0.6 m: halves equal throughout; pressure negative from tau_d = -0.36; D = 4, set B, form C, k_F,total = 0.3 m: halves unequal from tau_d = -0.82, one half empty from -0.47; pressure negative from tau_d = 0.06; D = 4, set B, form C, k_F,total = 0.6 m: halves unequal from tau_d = -1.98, one half empty from -1.62; pressure positive throughout; D = 4, set B, form Q, k_F,total = 0.3 m: halves equal throughout; pressure negative from tau_d = 0.51; D = 4, set B, form Q, k_F,total = 0.6 m: halves equal throughout; pressure negative from tau_d = -0.49; D = 3, set A, form C, k_F,total = 0.3 m: halves unequal from tau_d = -0.92, one half empty from -0.72; pressure negative from tau_d = -0.03; D = 3, set A, form C, k_F,total = 0.6 m: halves unequal from tau_d = -1.48, one half empty from -1.27; pressure positive throughout; D = 3, set A, form Q, k_F,total = 0.3 m: halves equal throughout; pressure negative from tau_d = 0.21; D = 3, set A, form Q, k_F,total = 0.6 m: halves equal throughout; pressure negative from tau_d = -0.21; D = 3, set B, form C, k_F,total = 0.3 m: halves unequal from tau_d = -1.02, one half empty from -0.88; pressure negative from tau_d = -0.12; D = 3, set B, form C, k_F,total = 0.6 m: halves unequal from tau_d = -1.52, one half empty from -1.32; pressure positive throughout; D = 3, set B, form Q, k_F,total = 0.3 m: halves equal throughout; pressure negative from tau_d = 0.10; D = 3, set B, form Q, k_F,total = 0.6 m: halves equal throughout; pressure negative from tau_d = -0.14

[PASS] fm-F6 (numerical) the Kohn-Sham method and the CJT effective potential agree for the two halves, in both forms: for 80 states (two dimensions, two sets, two forms, strengths 1.5 and 3 times the respective binding threshold, total densities of the Fermi momenta 0.3 to 1.1 m; 40 with equal halves, 17 with unequal halves, 23 with one half filled) the stationary point of Omega(M_+, M_-, V_+, V_-; mu) at the chemical potential of the Kohn-Sham state, found by Newton's method from displaced start values, has the same two masses, the same density and energy density, and -Omega equals the Kohn-Sham pressure (sum of the deviations at most 4.9e-11; residuals of the four stationarity conditions at most 1.4e-12). In the states with unequal halves the two chemical potentials agree to 6.7e-16 (relative): the polarization is the equilibrium one. In all 23 states with one half filled the lowest state of the empty half lies above the chemical potential

[PASS] fm-F7 (numerical) the vacuum of the two forms with the filled states of negative energy (cutoff Lambda). The gap equations of the vacuum are (M_+ + M_-)/2 - m = T_S (s_+ + s_-) and (M_+ - M_-)/2 = D_S (s_+ - s_-) with s_+- = -(NU/2) Int^Lambda M_+-/w. (a) Without mass term (m = 0) and for an attractive scalar coupling (sets A and B: G_S^HF < 0 < G_Sc) a field acquires a mass when |G_S^HF| exceeds the critical coupling |G_c| (D = 4: G_c Lambda^3 = -29.608813, D = 3: G_c Lambda^2 = -4.934802), in both forms at the same strength of the couplings: no mass 2 per cent below it, a mass 2 per cent above it. In the quantized-sector form the two halves acquire the SAME mass (M_+ = M_-: the mass term of the positive Lagrangian); in the covariant form they acquire masses of OPPOSITE sign (M_+ = -M_-: the mass term -m psitilde psi of the covariant Lagrangian), verified to 1e-7. (b) With the bare mass 0.05 Lambda the quantized-sector form passes smoothly through the critical strength (the two masses stay equal at all 101 clock readings), whereas the covariant form has a sharp transition at which the two masses separate: D = 4, set A: tau_d = 0.0091 (linear criterion 0.0091); D = 4, set B: tau_d = 0.0094 (linear criterion 0.0094); D = 3, set A: tau_d = 0.0194 (linear criterion 0.0194); D = 3, set B: tau_d = 0.0199 (linear criterion 0.0199); the strength of the transition agrees with the linear criterion 1 + D_S NU Int^Lambda d^Dk/(2 pi)^D k^2/w^3 = 0 at the common mass to 2.1e-10

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[PASS] fm-F8  (numerical) bare Dirac mass, self-interaction and Vorton-Scalar together, in the two
 ↪  forms (D = 3, no-sea approximation): the smallest C at which matter is self-bound, for f_D =
 ↪  0, 0.25, 0.5 and the contact couplings g_0 = -kappa_0 C, g_1 = +kappa_1 C. Without
 ↪  self-interaction both forms give the threshold of the model with the Vorton-Scalar alone, C =
 ↪  3.70110 and 3.70110 against 3 pi^2/8 = 3.70110. In 18 of the 18 cases the threshold is that of
 ↪  matter of one kind (the smaller of the thresholds of matter with equal halves and of matter
 ↪  with one half filled, functional of module kohn_sham), confirmed by the functional of two
 ↪  halves minimized over density, field and polarization: at 1.001 times the threshold its lowest
 ↪  state is bound (E/N - m <= -1.6e-4 m) and is of the expected kind, at 0.999 times the
 ↪  threshold no state is bound (E/N - m >= 1.6e-4 m); in 0 cases the threshold was found by
 ↪  bisection with the functional of two halves (a bound state with unequal halves). With
 ↪  self-interaction: f_D = 0, kappa_0 = 1, kappa_1 = 0, form C: C = 3.9399 (shift +0.063; equal
 ↪  halves, x = 0.000); f_D = 0, kappa_0 = 1, kappa_1 = 0, form Q: C = 3.7156 (shift +0.004; equal
 ↪  halves, x = 0.614); f_D = 0, kappa_0 = 1, kappa_1 = 0.1, form C: C = 3.9399 (shift +0.063;
 ↪  equal halves, x = 0.000); f_D = 0, kappa_0 = 1, kappa_1 = 0.1, form Q: C = 4.1527 (shift
 ↪  +0.115; equal halves, x = 0.652); f_D = 0.25, kappa_0 = 1, kappa_1 = 0, form C: C = 2.6336
 ↪  (shift -0.340; equal halves, x = 0.000); f_D = 0.25, kappa_0 = 1, kappa_1 = 0, form Q: C =
 ↪  2.4758 (shift -0.402; equal halves, x = 0.000); f_D = 0.25, kappa_0 = 1, kappa_1 = 0.1, form
 ↪  C: C = 2.6307 (shift -0.341; equal halves, x = 0.000); f_D = 0.25, kappa_0 = 1, kappa_1 = 0.1,
 ↪  form Q: C = 2.6992 (shift -0.316; equal halves, x = 0.000); f_D = 0.5, kappa_0 = 1, kappa_1 =
 ↪  0, form C: C = 2.1617 (shift -0.538; equal halves, x = 0.000); f_D = 0.5, kappa_0 = 1, kappa_1
 ↪  = 0, form Q: C = 1.8252 (shift -0.707; equal halves, x = 0.000); f_D = 0.5, kappa_0 = 1,
 ↪  kappa_1 = 0.1, form C: C = 2.1559 (shift -0.540; equal halves, x = 0.000); f_D = 0.5, kappa_0
 ↪  = 1, kappa_1 = 0.1, form Q: C = 1.9335 (shift -0.649; equal halves, x = 0.000)
[PASS] fm-F9  (numerical) a scalar coupling of the opposite sign, g_0 > 0 alone. In the covariant
 ↪  form particles of different halves attract each other; the coupling of the square of the total
 ↪  density is T_V = G_Vc = -g_0/16 < 0, and the energy per particle of uniform matter falls
 ↪  without bound as the density grows: the contact model then has no saturated state without a
 ↪  cutoff. In the quantized-sector form the same coupling repels (T_S = 15 g_0/16 > 0) and the
 ↪  energy per particle rises with the density. D = 4, g_0 = +100 m^-3, form C: E/N - m = +0.089,
 ↪  -1.518, -35.819, -641.710 m at k_F,total = 1, 2, 4, 8 m; D = 4, g_0 = +100 m^-3, form Q: E/N -
 ↪  m = +1.095, +15.250, +243.184, +3890.737 m at k_F,total = 1, 2, 4, 8 m; D = 3, g_0 = +15 m^-2,
 ↪  form C: E/N - m = +0.158, -0.054, -5.220, -58.160 m at k_F,total = 1, 2, 4, 8 m; D = 3, g_0 =
 ↪  +15 m^-2, form Q: E/N - m = +0.929, +6.205, +48.701, +389.107 m at k_F,total = 1, 2, 4, 8 m
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Appendix C. The symbols of this document

symbol	name in programs and data files	meaning
$x0, \dots, x7$	`x0` .. `x7`	the eight coordinates: $x0$ hidden space coordinate, $x1, x2, x3$ ordinary space, $x4$ ordinary time, $x5, x6, x7$ the extra coordinates (time-like on 4+4)
g55, g66, g77; gsum	`g55` `g66` `g77` `gsum`	the rates of deflation of $x5, x6, x7$ and their sum
V_d	`Vd`	proper volume of a unit coordinate cell of $x5, x6, x7$
τ_d	`tau_d`	the deflation clock, Eq. (1.1); the couplings of the quartic term grow as e^{τ_d}
γ^A ($= \gamma^A_{1644}$)	`gamma1644`	the eight real Dirac matrices of the 4+4 spacetime, Eq. (1.2)
C^-	`Cm1644`	$\gamma^0\gamma^1\gamma^2\gamma^3$; defines the covariant adjoint $\tilde{\psi} = \psi^\dagger C^-$
β, α^j	`beta1644`	$-i\gamma^4$ and $-\gamma^4\gamma^j$; $\bar{\psi} = \psi^\dagger\beta$
W ($= W_{1644}$)	`W`	$i\,C^-\gamma^4$; its eigenvalues ± 1 define the two halves of the field
χ	`omega1644`	the chirality matrix, the product of the eight Dirac matrices
h	—	the Hamiltonian of one particle of the quantized sector, Eq. (2.3)
$\psi_\pm$	—	the two halves of the field, $W\psi_\pm = \pm\psi_\pm$
K, r	—	a list of different indices, and its rank

symbol	name in programs and data files	meaning
Θ^K, Θ_K, ζ_r, ε_K, η_K	`Theta`	the 256 matrices of form C, their inverses, phases and signs, Eqs. (3.2), (3.3)
γ^A_{44q}, Θ^K_Q, ζ^Q_r, ε^Q_K	`gamma44q`	the matrices of form Q, Eqs. (3.6), (3.7)
I_r, I^{C}_r, I^{Q}_r	`I_r`	the invariant of rank r, in general and in the two forms
g_r	`g_0` .. `g_8`	the nine couplings of the self-interaction
g^{eff}_0, g^{eff}_1, g^{eff}_7, g^{eff}_8	`g_0eff` ..	the four effective couplings, Eq. (4.5)
`F`	`F_r0` ..	the Fierz matrix
z_K	—	the phase of Eq. (4.1)
$n_\pm$, $n_{S\pm}$	—	number and scalar densities of the two halves
n, n_S, n_W, n_{SW}	`rho` `rho_S` `rho_W` `rho_SW`	their sums and differences, Eq. (5.2)
G^{HF}_S, G^{HF}_V, G^{c}_S, G^{c}_V	`G_0` `G_1` `G_S0` `G_V0`	the four mean-field combinations, Eq. (5.4)
T_V, T_S, D_V, D_S	`T_V` `T_S` `D_V` `D_S`	couplings of the total densities and of the differences, Eq. (5.3)
a_S, a_V, b_S, b_V	`a_S` `a_V` `b_S` `b_V`	couplings between particles of the same half and of different halves, Eq. (5.6)
g	`g`	the strength of the couplings of the sets A and B
$M_\pm$, $V_\pm$, $\mu_\pm$, $k_\pm$	`M_plus` `M_minus` `mu_plus` `mu_minus`	Kohn–Sham masses, energy shifts, chemical potentials and Fermi momenta of the two halves
$k_{F,\mathrm{tot}}$	`kF_total`	the Fermi momentum that the total density would have with equal halves
pol_W	`pol`	the polarization, Eq. (5.12)
ε, p, Π	`eps` `p` `Pi`	energy density, pressure, interaction pressure
R	`response`	the response matrix of the gap equations, Eq. (5.13); the data files give its largest eigenvalue
D	`D`	the number of space dimensions in which the particles move (4 or 3)
Ω, Ω_1, μ_S, d_S, v_T, v_D	`Omega`	the effective potential of two halves, Eqs. (5.14), (5.15)
Λ, G_c, g_{crit}	`Gc` `g_crit`	cutoff of the momentum, critical coupling (6.2), critical strength
Λ_{max}	`cutoff_max`	the largest cutoff for which the vacuum is stable at a given strength
X, κ	—	matrix of a pair operator, and pair density
G_{pair}	—	the coupling of a pair of two particles at rest
F_0, F_1, F_7, F_8; F_S, F_P	—	pairing amplitudes of the contact channels and of the two modes of the scalar, Eqs. (7.2), (7.5)
s	`ct_sign`	the sign with which the contact couplings enter the pairing potential, Eq. (7.4)
g_{VS}, m_s, m_ϑ	`g` `m_s`	coupling of the Vorton-Scalar, masses of its radial and of its phase mode
m_{ct}	—	the range parameter of the contact couplings in the gap equation
Δ, `W`, ξ, K (upper limit)	`gap`	the gap, the pairing kernel, the energy from the Fermi surface, the upper limit of the momentum integral
`w`, J, `s`, σ, U	`uv`	the kernel at large momenta, the integral (7.7), the exponents, the ultraviolet strength (7.8); the data files give $U/4\pi^2$
I_S, I_V	—	the two integrals of Eq. (7.9)
P_E, $U(1)_E$	—	the generator $\propto \gamma^0\gamma^5\gamma^6\gamma^7$ and its phase symmetry
`a`, `b`; H_{a}, H_{b}	`a` `b` `H_a` `H_b`	scale factors of the four space-like and of the three extra coordinates, and their rates, Eq. (10.1)
K_{pair}, $E^{(k)\mu}{}_\nu$, α_k, $\bar\Lambda$, κ_8	—	pair curvatures (10.2), Lovelock tensors (10.4), their couplings, cosmological term and gravitational coupling of Eq. (10.3)
loading	`loading`	$\kappa_8\,\varepsilon(0)/(3\,\mathrm{g55}^2)$
$SO(3)_T$	—	the rotations of $x5, x6, x7$ among themselves
ϕ, v_{VS}, x, C, f_D, m_D, m_Y, κ_0, κ_1	`x` `C_bind` `f_D` `kappa_0` `kappa_1`	the scalar field, its vacuum value, their ratio, the coupling $g^2_{VS}m^2/m^2_s$, the fraction of bare mass, the bare and the Yukawa mass, the contact couplings in units of C